

Parameterized Penguins Report

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```
library("webshot2")
```

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1 Init

1.1 Summary software

- **Operating System:** Ubuntu 22.04.5 LTS
- **Platform info:** x86_64-pc-linux-gnu
- **R Info:** R version 4.4.3 (2025-02-28) – “Trophy Case”
- **Rscience Version:** ~~XXX~~ agricolae 1.3.7
- **UTC Time:** 2025-10-31 09:57:43 UTC
- **System/Computer Time:** 2025-10-31 10:57:43 CET
- **Inferred Timezone Location:** Europe/Rome
(R can only reliably determine the timezone, not the physical city/country.)

1.2 User Inputs

- **Rscience Tool:** General Linnear Model, Fixed Effect, anova 1 way
- **Rscience Tool Script:** Script XXX-001
- **Dataset file name:** mtcars
- **Response Variable:** mpg
- **Factor:** cyl
- **Alpha value:** 0.05

2 AutoReport

2.1 Report (Automated Statistical Assistant)

- **Dataset file name:** jaja
- **Response Variable:** mpg
- **Factor:** cyl
- **Alpha value:** 0.05

The null hypothesis of normal distribution of residuals is not rejected.

The hypothesis of homogeneity of variances (homoscedasticity) is rejected.

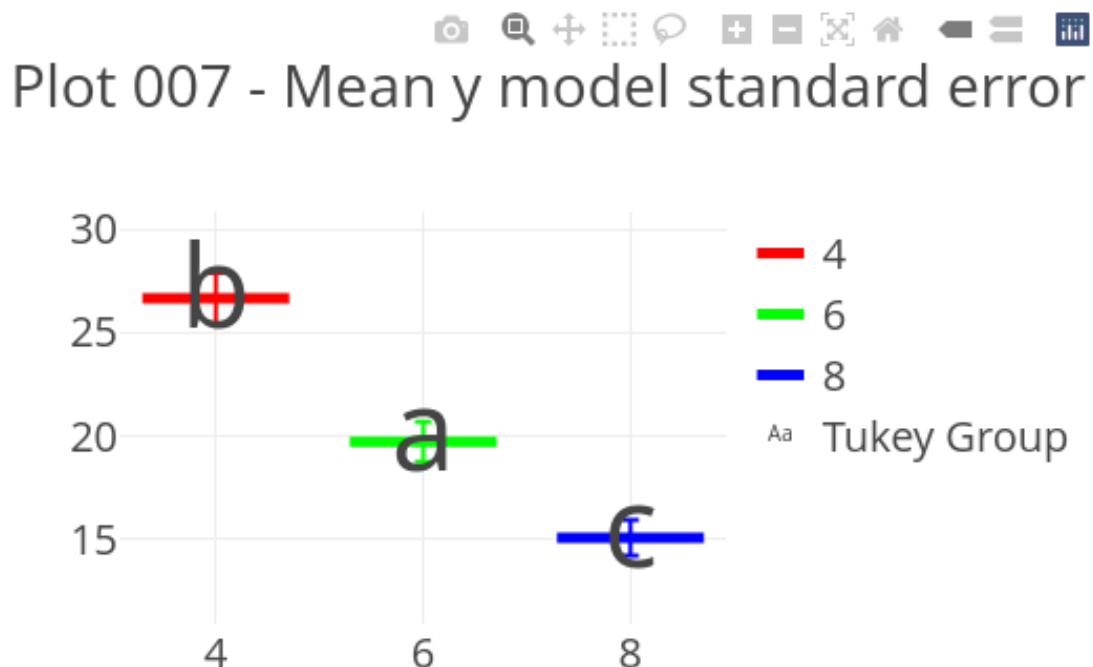
Not all model assumptions are met, so it is NOT valid to draw conclusions from the ANOVA test.

Regardless of the p-value obtained in ANOVA, it is not valid to draw conclusions.

2.1.a 1) Summary Anova

test	aim	variable	p_value	alpha_value	Decision
Shapiro-Wilk test	Normality	residuals	0.5176650	0.05	Ho no rejected
Bartlett test	Homo-geneity	residuals	0.0150452	0.05	Ho Rejected
Anova 1 way	Mean	mpg	0.0000000	0.05	Ho Rejected

2.1.b 2) Plot means and standard error from model



2.1.c 3) Tukey table

order	level	mean	group
1	4	26.66364	a
2	6	19.74286	b
3	8	15.10000	c

3 Classroom (ANOVA)

The most important statistical output from Anova as R formatted objects.

3.1 Model

The one-way ANOVA model can be expressed in two equivalent forms:

3.1.a Form 1: Treatment Effects Model

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

3.1.b Form 2: Group Means Model

$$Y_{ij} = \mu_i + \varepsilon_{ij}$$

3.2 Model Components

Where: - Y_{ij} = j -th observation in the i -th group - μ = Overall mean (grand mean) - τ_i = Effect of the i -th treatment - μ_i = Mean of the i -th group - ε_{ij} = Random error term

3.3 Key Relationship

The relationship between the two formulations is given by:

$$\mu_i = \mu + \tau_i$$

3.4 Model Parameters

- k = number of groups or treatments
- μ_i = population mean of group i
- $i = 1, 2, \dots, k$ (group index)
- $j = 1, 2, \dots, n_i$ (observation index within each group)
- n_i = number of observations in group i

3.5 Error Term Specification

The random error term follows a normal distribution:

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

Where: - σ^2 = common variance across all groups

3.6 Summary One-Way ANOVA Model

Every general linear model involves estimating parameters for a polynomial in order to predict a quantitative response variable with special requirements of normality, homogeneity of variances, and independence of residuals.

The fixed effects general linear model of One-Way ANOVA involves:

- The response variable or dependent variable must be quantitative.
- The predictor variable must be a qualitative independent variable with at least 2 levels. Particularly in ANOVA models, this variable is called a Factor.

- The expected component of the model arises from estimating zero-degree polynomials, one for each factor level. Estimating the means of each factor level involves estimating the intercept term of a zero-degree polynomial.
- The error component must have a normal distribution, zero mean, homogeneous variance, and the residuals must be independent.
- The normality test is performed using the Shapiro-Wilk test.
- Homogeneity is formally tested using Bartlett's test.
- Independence concerns the errors. Errors are independent if the statistical units are independent. Statistical units are independent if the design is carried out under this concept. Independence is a required characteristic in the design and cannot be tested in any way.

3.7 Hypothesis

3.8 Statistical Hypotheses

3.8.a Form 1: Treatment Effects (τ_i)

Hypothesis Type	Mathematical Formulation	Verbal Description
Null Hypothesis (H_0)	$H_0 : \tau_1 = \tau_2 = \dots = \tau_k = 0$	All treatment effects are zero
Alternative Hypothesis (H_1)	$H_1 : \exists i \in \{1, 2, \dots, k\} \mid \tau_i \neq 0$	At least one treatment effect is different from zero

3.8.b Form 2: Group Means (μ_i)

Hypothesis Type	Mathematical Formulation	Verbal Description
Null Hypothesis (H_0)	$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$	All group means are equal
Alternative Hypothesis (H_1)	$H_1 : \exists i, j \in \{1, 2, \dots, k\} \mid i \neq j \wedge \mu_i \neq \mu_j$	At least one group mean is different

3.9 Assumption Tests for ANOVA Residuals

3.9.a Normality Test (e.g., Shapiro-Wilk)

Hypothesis Type	Mathematical Formulation	Verbal Description
Null Hypothesis (H_0)	$H_0 : \varepsilon_{ij} \sim N(0, \sigma^2)$	The residuals are normally distributed
Alternative Hypothesis (H_1)	$H_1 : \varepsilon_{ij} \not\sim N(0, \sigma^2)$	The residuals are not normally distributed

3.9.b Homogeneity of Variances Test (e.g., Bartlett's Test)

Hypothesis Type	Mathematical Formulation	Verbal Description
Null Hypothesis (H_0)	$H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$	All group variances are equal
Alternative Hypothesis (H_1)	$H_1 : \exists i, j \in \{1, 2, \dots, k\} \mid i \neq j \wedge \sigma_i^2 \neq \sigma_j^2$	At least one group variance is different

3.10 Decision Rule

The decision to reject or not reject the null hypothesis is based on the p-value and the significance level α :

If $p < \alpha \Rightarrow$ Reject H_0

If $p \geq \alpha \Rightarrow$ Fail to reject H_0

Where: - p = p-value from the statistical test - α = significance level (typically 0.05) - k = number of groups or treatments - σ_i^2 = variance of group i - $\exists$ = “there exists” (existential quantifier)

3.11 User election

3.11.a Selected variable and rol

order	var_name	var_number	var_letter	var_role	doble_reference
1	mpg	1	A	RV	RV(mpg)
2	cyl	2	B	FACTOR	FACTOR(cyl)

3.11.b Alpha and confidence values

order	detail	probability	percentaje
1	alpha value	0.05	5%
2	confidence value	0.95	95%

3.11.c Amount of data to analysis

object	n_col	n_row
my_dataset	11	32
minibase	2	32

3.12 Anova Table

3.12.a 1) Anova Table (Class Room format)

This is the table format detailed in the bibliography.

	Sources of Variation	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cyl	Factor	2	824.7846	412.39229	39.69752	0
Residuals	Error	29	301.2626	10.38837	NA	NA
Total	Total	31	1126.0472	36.32410	NA	NA

3.12.b 2) Anova Table (R format)

The original R output does not contain the Source of Variation column, and it does not detail the row for the total source of variation.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cyl	2	824.7846	412.39229	39.69752	0
Residuals	29	301.2626	10.38837	NA	NA

3.13 Tukey Test

3.13.a 1) Tukey Test (Class Room format)

This is the table format detailed in the bibliography.

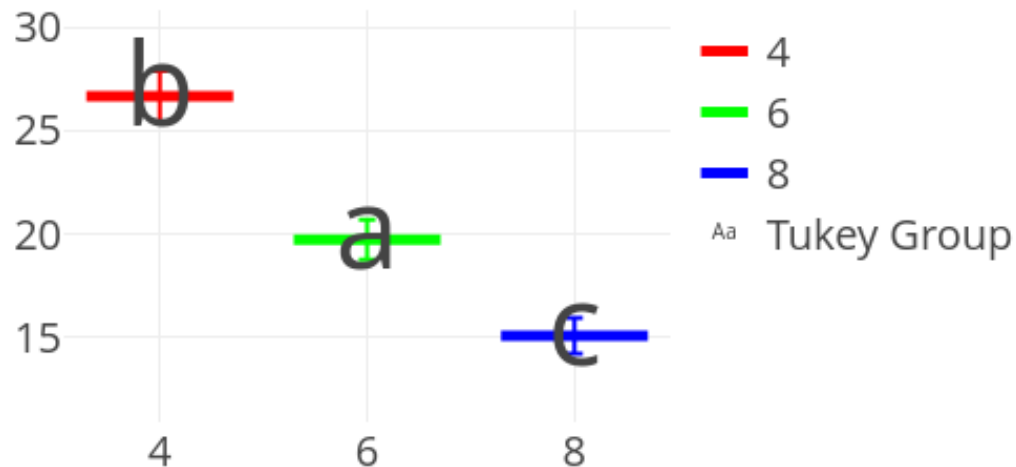
order	level	mean	group
1	4	26.66364	a
2	6	19.74286	b
3	8	15.10000	c

3.13.b 2) Special plot for Tukey test

The means of each factor level are detailed with the standard error derived from the model. The statistical groups from the Tukey test are detailed for each factor level.

file:///tmp/RtmpooteVR/file58cd309cad2a/widget58cd7758d012.html screenshot completed

Plot 007 - Mean y model standard error



3.13.c 3) Full info from Tukey (R format)

The Tukey table is only one part of all the information contained within the Tukey test.

The Tukey test contains more details that may be relevant, such as the honest significant difference (HSD).

```
$statistics
  MSerror Df      Mean      CV
10.38837 29 20.09062 16.0428

$parameters
  test name.t ntr StudentizedRange alpha
Tukey   cyl   3          3.492609  0.05

$means
      mpg      std r      se Min Max  Q25  Q50  Q75
4 26.66364 4.509828 11 0.9718008 21.4 33.9 22.80 26.0 30.40
6 19.74286 1.453567  7 1.2182168 17.8 21.4 18.65 19.7 21.00
8 15.10000 2.560048 14 0.8614094 10.4 19.2 14.40 15.2 16.25

$comparison
NULL

$groups
      mpg groups
4 26.66364    a
6 19.74286    b
8 15.10000    c
```

```
attr(,"class")
[1] "group"
```

3.13.d 4) Pairs comparison from Tukey test (R format)

Internally, the Tukey test performs t-tests between all possible pairs of factor levels.

In each t-test, a p-value is obtained with which decisions are made. All pairs are detailed below.

```
$statistics
  MSerror Df      Mean      CV
10.38837 29 20.09062 16.0428

$parameters
  test name.t ntr StudentizedRange alpha
Tukey   cyl   3           3.492609  0.05

$means
      mpg      std  r      se Min  Max  Q25  Q50  Q75
4 26.66364 4.509828 11 0.9718008 21.4 33.9 22.80 26.0 30.40
6 19.74286 1.453567  7 1.2182168 17.8 21.4 18.65 19.7 21.00
8 15.10000 2.560048 14 0.8614094 10.4 19.2 14.40 15.2 16.25

$comparison
      difference pvalue signif.      LCL      UCL
4 - 6    6.920779 0.0001     *** 3.330308 10.511251
4 - 8   11.563636 0.0000     *** 7.973165 15.154108
6 - 8    4.642857 0.0092     **  1.052385  8.233329

$groups
NULL

attr(,"class")
[1] "group"
```

3.14 Requeriments

3.15 1) Requeriment - Normality test - Residuals (R format)

```
$test_residuals_normality

  Shapiro-Wilk normality test

data:  minibase_mod$residuals
W = 0.97065, p-value = 0.5177
```

3.16 2) Requeriment - Homogeneity test - Residuals (R format)

```
$test_residuals_homogeneity

  Bartlett test of homogeneity of variances
```



```
data: residuals by cyl
Bartlett's K-squared = 8.3934, df = 2, p-value = 0.01505
```

3.17 3) Estimated variances - Residuals

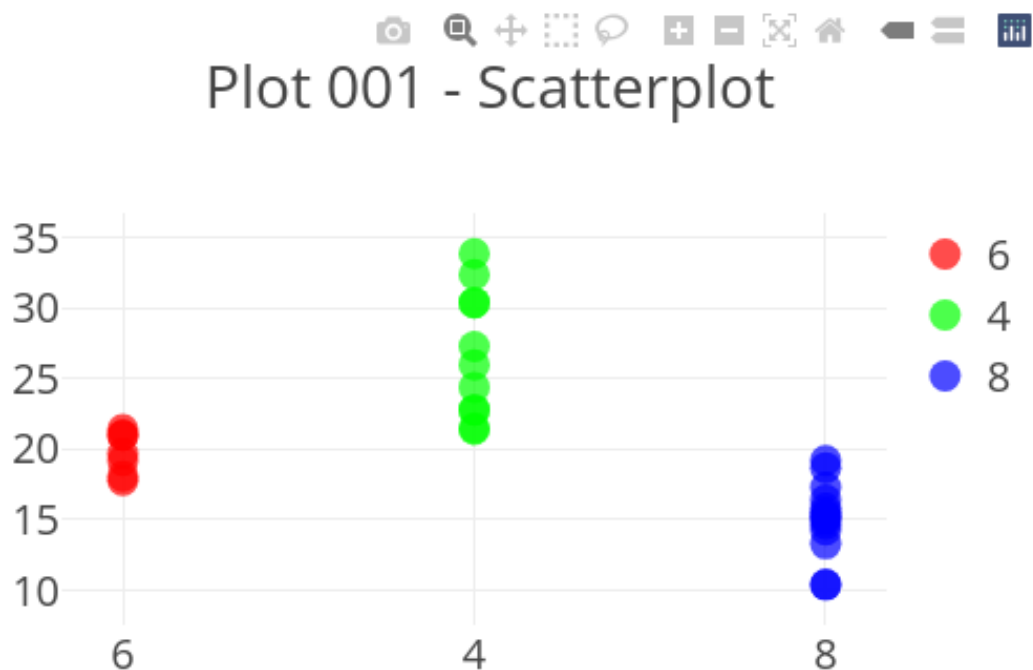
or- der	level	n	model_error_ var_MSE	model_er- ror_sd	model_er- ror_se	raw_er- ror_se
1	6	7	10.38837	3.223099	1.2182168	1.2182168
2	4	11	10.38837	3.223099	0.9718008	0.9718008
3	8	14	10.38837	3.223099	0.8614094	0.8614094

order	level	n	raw_error_var	raw_error_sd
1	6	7	2.112857	1.453567
2	4	11	20.338545	4.509828
3	8	14	6.553846	2.560048

Anova and Tukey use MSE from model. Bartlett use variance from raw error on each level. Only if there is homogeneity from raw error variances then is a good idea take desition from MSE.

3.18 Plots - Raw Data

```
file:///tmp/RtmpooteVR/file58cd2099f215/widget58cd7546a1d4.html screenshot
completed
```

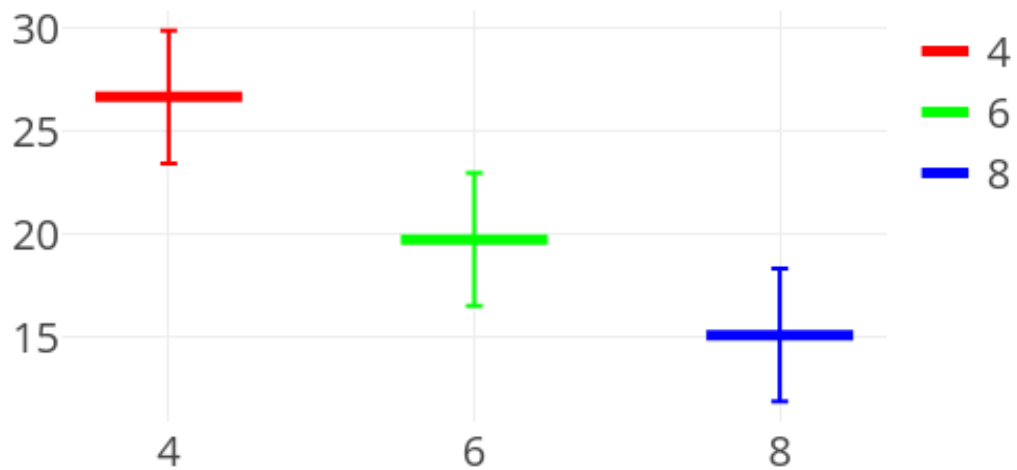


	order	level	n	mean	min	max	sd	var
6	1	6	7	19.74286	17.8	21.4	1.453567	2.112857
4	2	4	11	26.66364	21.4	33.9	4.509828	20.338545
8	3	8	14	15.10000	10.4	19.2	2.560048	6.553846

file:///tmp/RtmpooteVR/file58cd7dc00034/widget58cd19623794.html screenshot completed



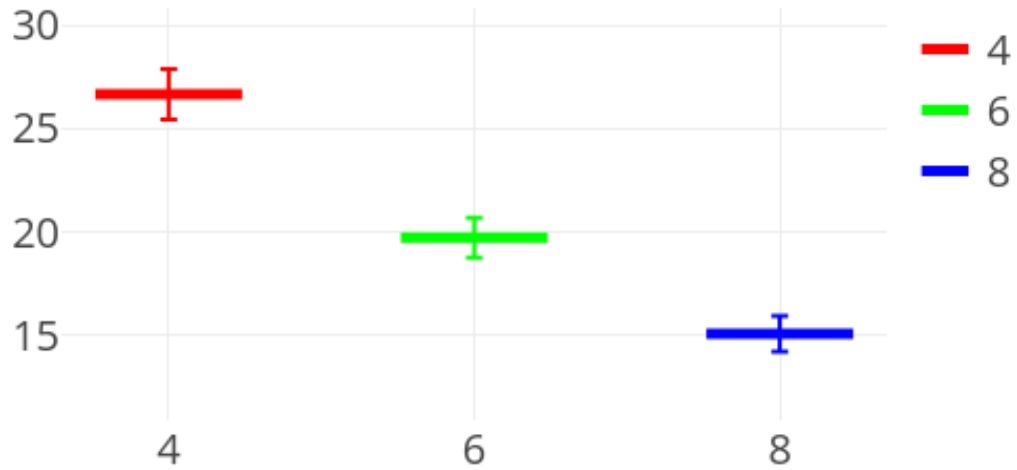
it 002 - Mean and model standard deviat



or- der	level	n	mean	model_error_sd	lower_limit	upper_lim- mit	color	
6	1	6	7	19.74286	3.223099	16.51976	22.96596	#FF0000
4	2	4	11	26.66364	3.223099	23.44054	29.88674	#00FF00
8	3	8	14	15.10000	3.223099	11.87690	18.32310	#0000FF

file:///tmp/RtmpooteVR/file58cd26bc7f1a/widget58cd7a6c70d0.html screenshot completed

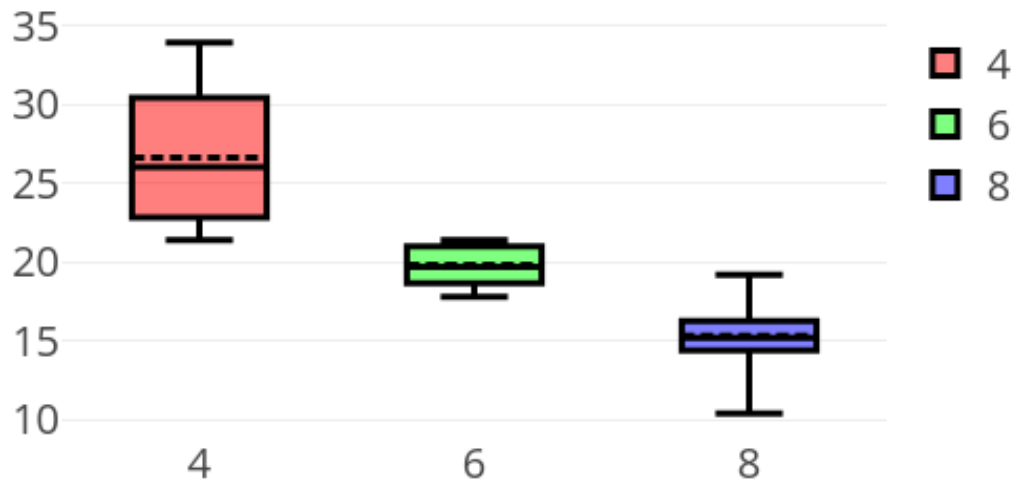
Plot 003 - Mean y model standard error



or- der	level	n	mean	model_error_se	lower_limit	upper_limit	color	
6	1	6	7	19.74286	1.2182168	18.52464	20.96107	#FF0000
4	2	4	11	26.66364	0.9718008	25.69184	27.63544	#00FF00
8	3	8	14	15.10000	0.8614094	14.23859	15.96141	#0000FF

file:///tmp/RtmpooteVR/file58cd40d2ccd1/widget58cd6e107d0.html screenshot completed

Plot 004 - Boxplot and means



	order	level	min	mean	Q1	median	Q3	max	n	color
6	1	6	17.8	19.74286	18.65	19.7	21.00	21.4	7	#FF0000
4	2	4	21.4	26.66364	22.80	26.0	30.40	33.9	11	#00FF00
8	3	8	10.4	15.10000	14.40	15.2	16.25	19.2	14	#0000FF

file:///tmp/RtmpooteVR/file58cd743d5074/widget58cd42804889.html screenshot completed

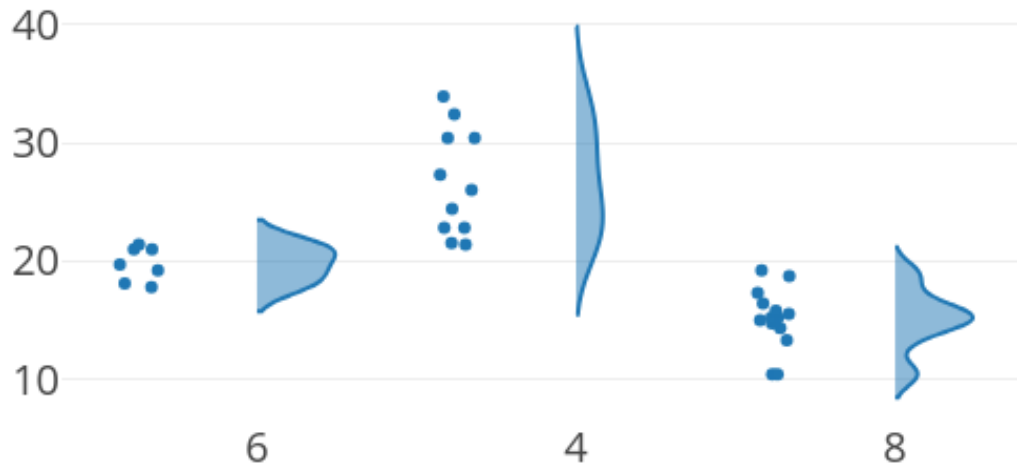
Plot 005 - Violinplot



	order	level	min	mean	Q1	median	Q3	max	n	color
6	1	6	17.8	19.74286	18.65	19.7	21.00	21.4	7	#FF0000
4	2	4	21.4	26.66364	22.80	26.0	30.40	33.9	11	#00FF00
8	3	8	10.4	15.10000	14.40	15.2	16.25	19.2	14	#0000FF

file:///tmp/RtmpooteVR/file58cd74ae711c/widget58cd568d164e.html screenshot completed

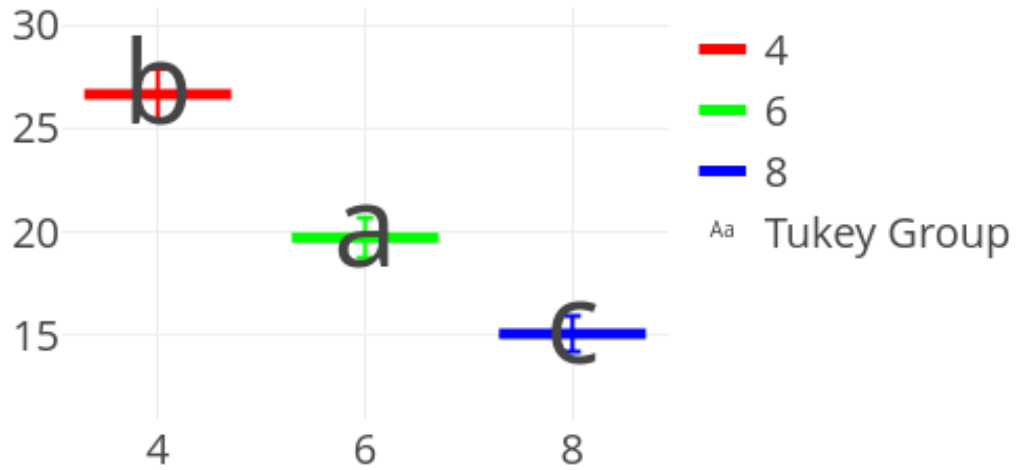
Plot 006 - Scatterplot + Jitter + Smoothed



	order	level	min	mean	Q1	median	Q3	max	n	color
6	1	6	17.8	19.74286	18.65	19.7	21.00	21.4	7	#FF0000
4	2	4	21.4	26.66364	22.80	26.0	30.40	33.9	11	#00FF00
8	3	8	10.4	15.10000	14.40	15.2	16.25	19.2	14	#0000FF

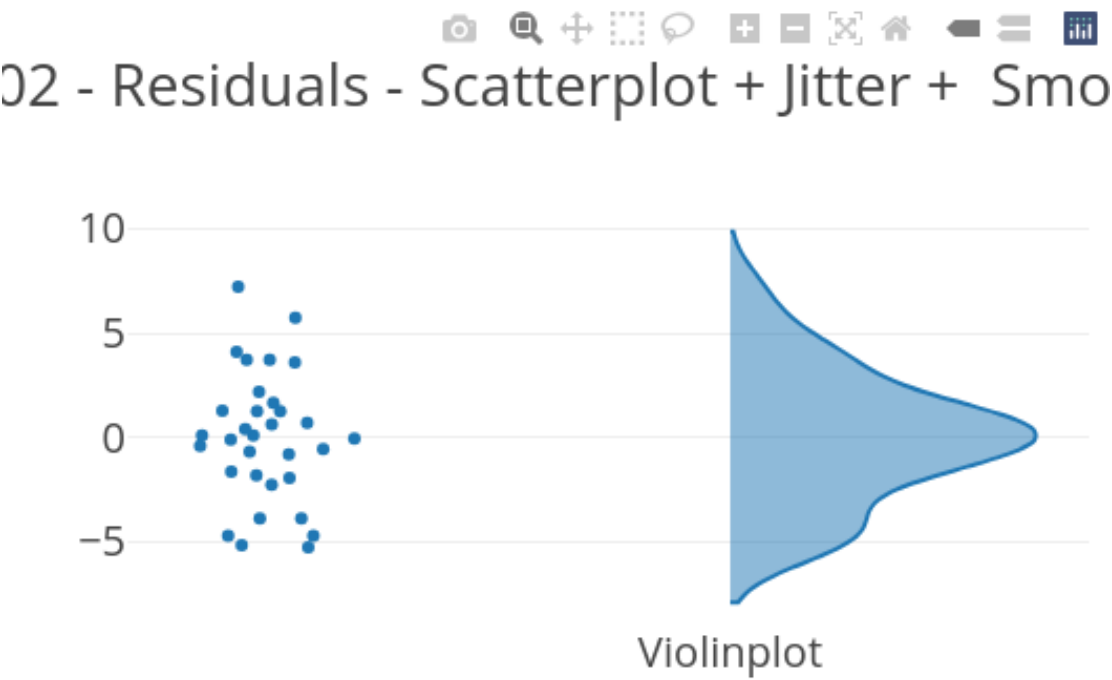
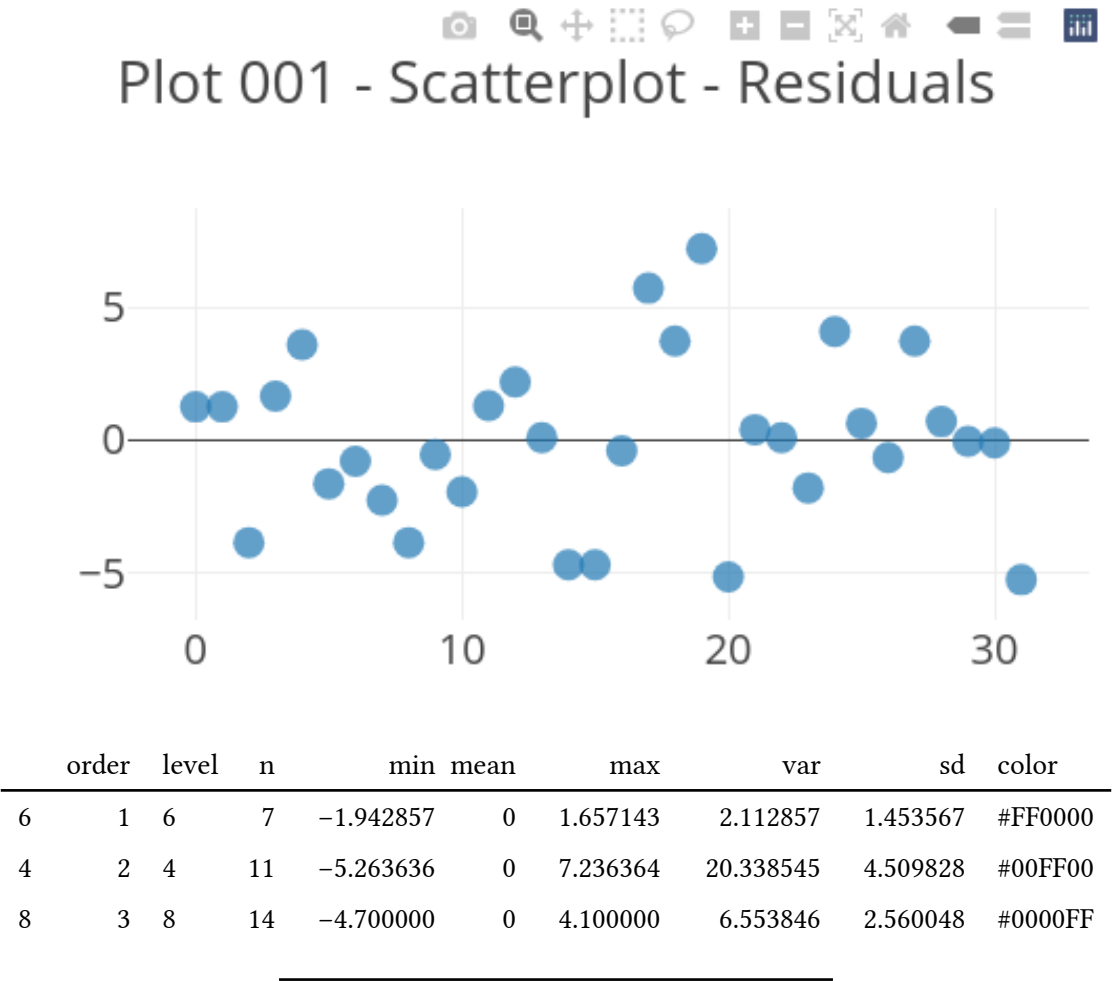
file:///tmp/RtmpooteVR/file58cda54a36a/widget58cd3327510d.html screenshot completed

Plot 007 - Mean y model standard error



or- der	level	n	mean	model_er- ror_se	low- er_limit	upper_lim- mit	color	group	
6	1	6	7	19.74286	1.2182168	18.52464	20.96107	#FF0000	a
4	2	4	11	26.66364	0.9718008	25.69184	27.63544	#00FF00	b
8	3	8	14	15.10000	0.8614094	14.23859	15.96141	#0000FF	c

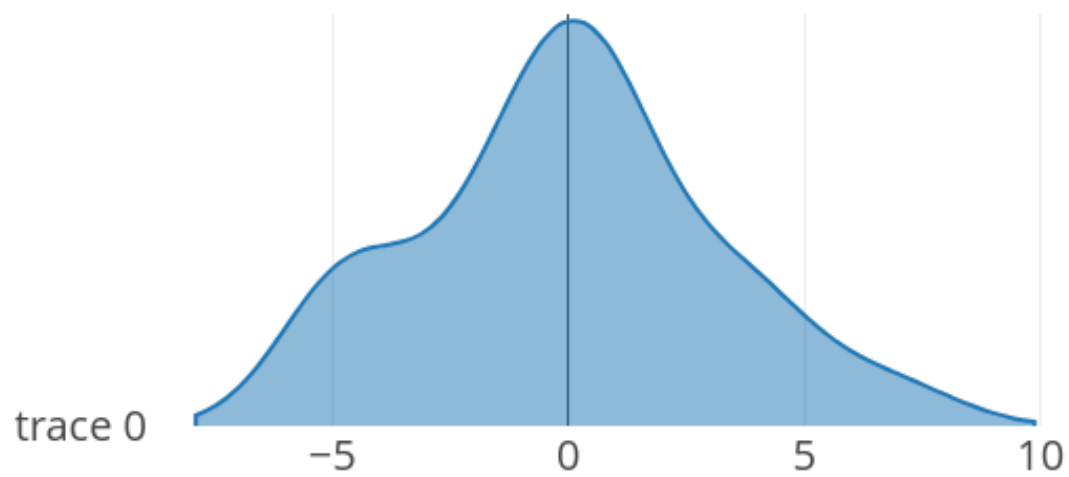
3.19 Plots - Residuals



	order	level	n	min	mean	max	var	sd	color
6	1	6	7	-1.942857	0	1.657143	2.112857	1.453567	#FF0000
4	2	4	11	-5.263636	0	7.236364	20.338545	4.509828	#00FF00
8	3	8	14	-4.700000	0	4.100000	6.553846	2.560048	#0000FF



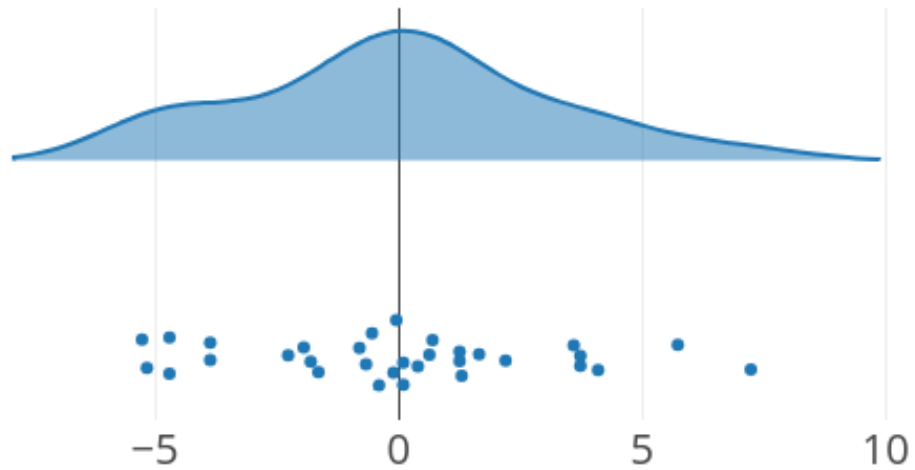
03 - Residuals - Scatterplot + Jitter + Smo



	order	level	n	min	mean	max	var	sd	color
6	1	6	7	-1.942857	0	1.657143	2.112857	1.453567	#FF0000
4	2	4	11	-5.263636	0	7.236364	20.338545	4.509828	#00FF00
8	3	8	14	-4.700000	0	4.100000	6.553846	2.560048	#0000FF



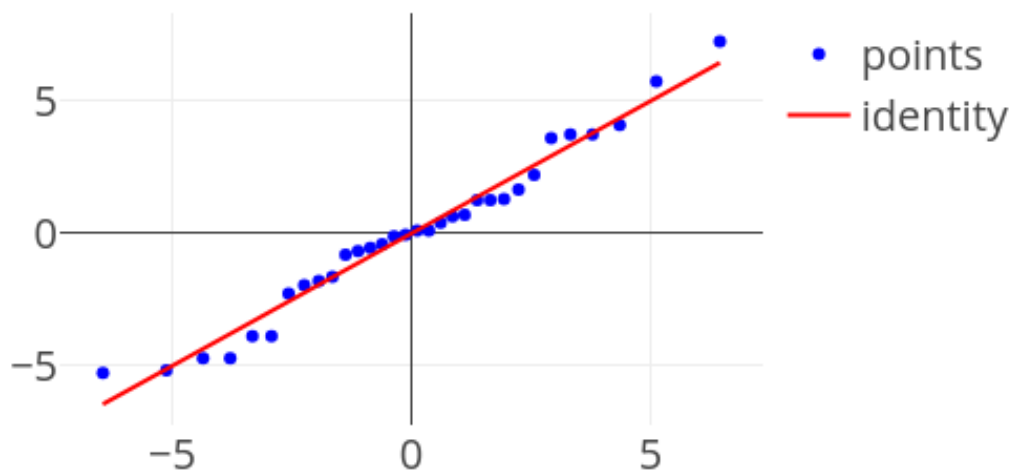
Plot 004 - Residuals



variable	n	min	mean	max	var	sd	model_error_var_MSE	model_error_sd
residuals	32	-5.263636	0.7236364	9.718148	3.117395		10.38837	3.223099

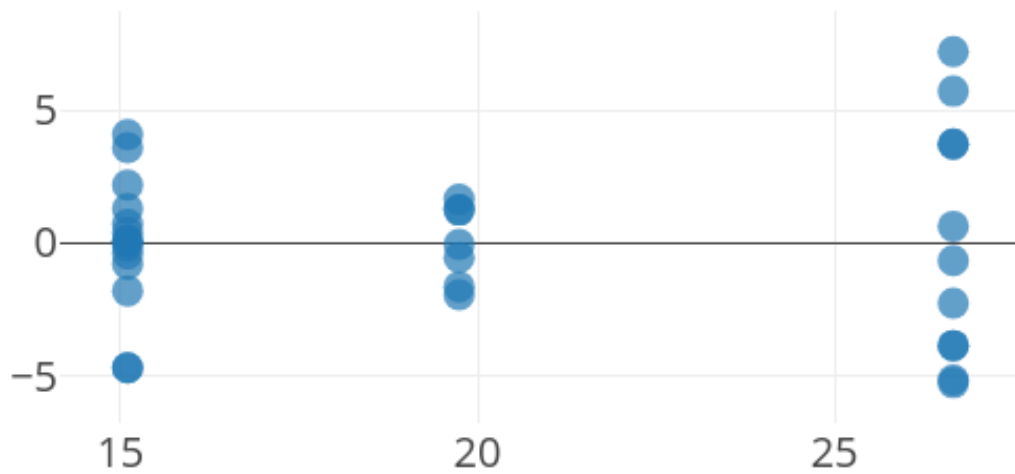


Plot 005 - QQ Plot Residuals



variable	n	min	mean	max	var	sd	model_error_var_MSE	model_error_sd
residuals	32	-5.263636	0	7.236364	9.718148	3.117395	10.38837	3.223099

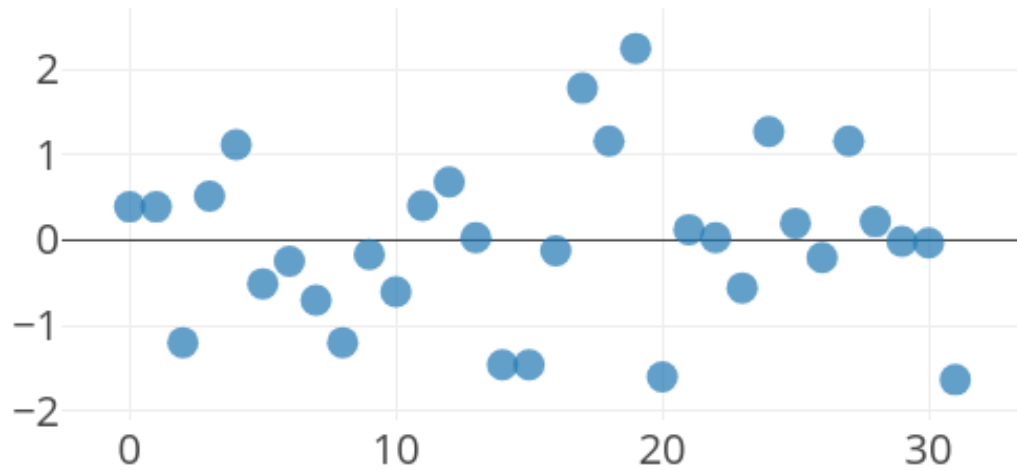
006 - Scatterplot - Residuals vs Fitted.val



	or- der	level	n	min	mean	max	var	sd	color
6	1	6	7	-0.6027917	0	0.5141459	0.2033869	0.4509843	#FF0000
4	2	4	11	-1.6330981	0	2.2451573	1.9578196	1.3992211	#00FF00
8	3	8	14	-1.4582240	0	1.2720678	0.6308833	0.7942816	#0000FF



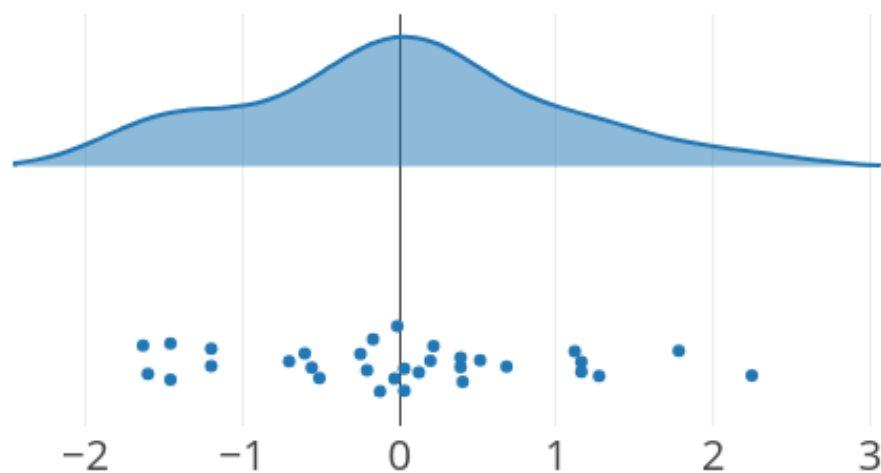
Plot 007 - Scatterplot - Studentized Residuals



	or- der	level	n	min	mean	max	var	sd	color
6	1	6	7	-0.6027917	0	0.5141459	0.2033869	0.4509843	#FF0000
4	2	4	11	-1.6330981	0	2.2451573	1.9578196	1.3992211	#00FF00
8	3	8	14	-1.4582240	0	1.2720678	0.6308833	0.7942816	#0000FF



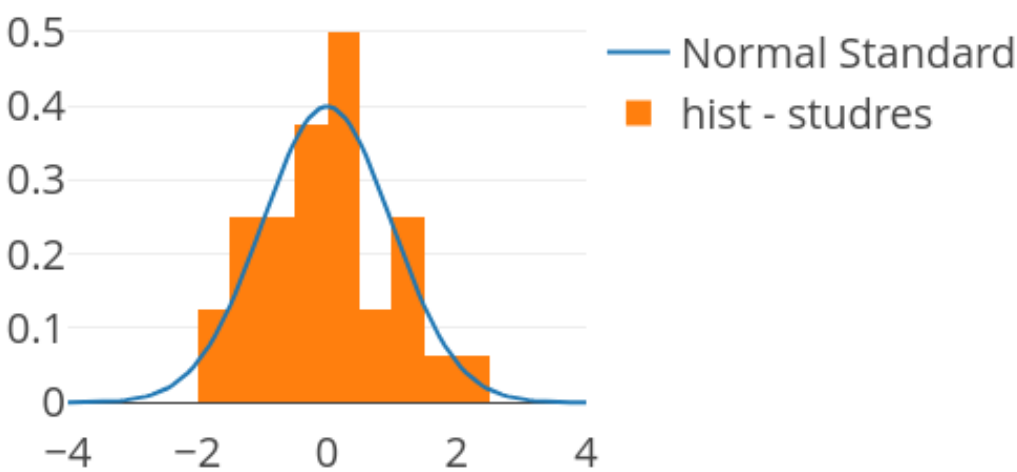
Plot 008 - Studentized Residuals



variable	n	min	mean	max	var	sd
studres	32	-1.633098	0	2.245157	0.9354839	0.9672042

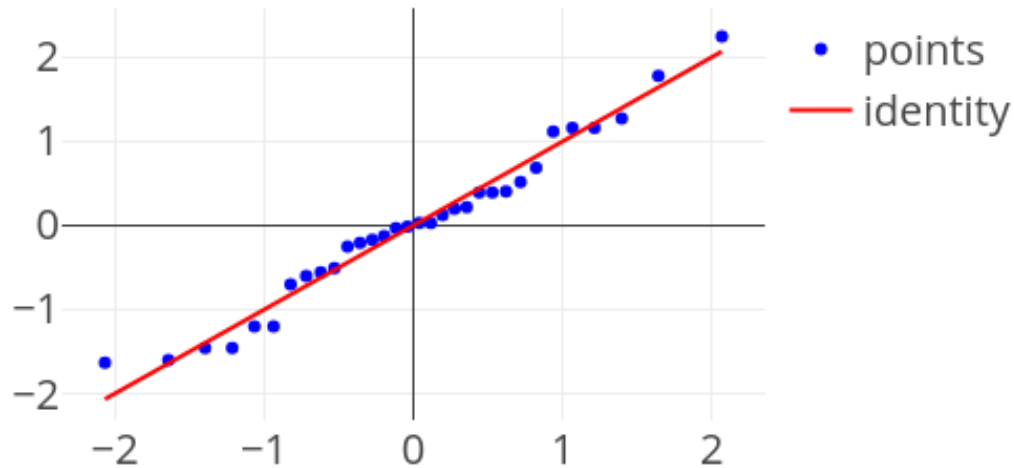


Plot 009 - Studres Distribution



variable	n	min	mean	max	var	sd
studres	32	-1.633098	0	2.245157	0.9354839	0.9672042

Plot 010 - QQ Plot - studres



variable	n	min	mean	max	var	sd
studres	32	-1.633098	0	2.245157	0.9354839	0.9672042

3.20 Decision Plan

3.21 Decision-Making Process in ANOVA

The ANOVA test is specifically performed to make decisions.

The decision-making process has a set of sequential steps, in a particular order.

- Step 01: Control over the response variable.
- Step 02: Verification of the normality requirement of the residuals.
- Step 03: Verification of the homogeneity of variances requirement.
- Step 04: Interpretation of the ANOVA test.
- Step 05: Interpretation of the Tukey test.

Haz clic en la imagen para ver en tamaño completo

3.22 Possible Analysis Cases

Decision-making in One-Way ANOVA involves 5 steps.

Each step has 2 possibilities: Yes or No.

This implies that for a given dataset, the analysis can lead to one of $2^5 = 32$ possibilities.

Fortunately, these 32 possibilities can be summarized into 6 analysis cases.

Any dataset falls into one of these cases, and this implies at least the possibility of providing a general statistical interpretation of what happens to that dataset.

3.23 Data Analysis (Step by step)

3.24 Decision-Making Process in ANOVA

The ANOVA test is specifically performed to make decisions.

The decision-making process has a set of sequential steps, in a particular order.

- Step 01: Control over the response variable.
- Step 02: Verification of the normality requirement of the residuals.
- Step 03: Verification of the homogeneity of variances requirement.
- Step 04: Interpretation of the ANOVA test.
- Step 05: Interpretation of the Tukey test.

3.24.a Step 01: Control over the response variable.

De la estadística descriptiva buscamos los valores mínimo y máximos de la variable respuesta par cada nivel del factor.

Esto es acción que permite detectar posibles errores de tipeo realizados en nuestro dataset. El control solo puede ser llevado a cabo en aquellos contextos en los cuales se tiene noción de lo que podría ser un valor imposible para la variable. Si se trabaja sin un marco de límite para la variable respuesta, deberá asumirse que los valores mínimos y máximos son válidos y se continúa con el siguiente paso.

	order	level	min	mean	Q1	median	Q3	max	n
6	1	6	17.8	19.74286	18.65	19.7	21.00	21.4	7
4	2	4	21.4	26.66364	22.80	26.0	30.40	33.9	11
8	3	8	10.4	15.10000	14.40	15.2	16.25	19.2	14

3.24.b Step 02: Verification of the normality requirement of the residuals.

```
$test_residuals_normality

Shapiro-Wilk normality test

data:  minibase_mod$residuals
W = 0.97065, p-value = 0.5177
```

The null hypothesis of normal distribution of residuals is not rejected.

3.24.c Step 03: Verification of the homogeneity of variances requirement.

```
$test_residuals_homogeneity

Bartlett test of homogeneity of variances

data:  residuals by cyl
Bartlett's K-squared = 8.3934, df = 2, p-value = 0.01505
```

The hypothesis of homogeneity of variances (homoscedasticity) is rejected.

3.24.d Step 04: Interpretation of the ANOVA test.

	Sources of Variation	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cyl	Factor	2	824.7846	412.39229	39.69752	0
Residuals	Error	29	301.2626	10.38837	NA	NA
Total	Total	31	1126.0472	36.32410	NA	NA

Not all model assumptions are met, so it is NOT valid to draw conclusions from the ANOVA test.

Regardless of the p-value obtained in ANOVA, it is not valid to draw conclusions.

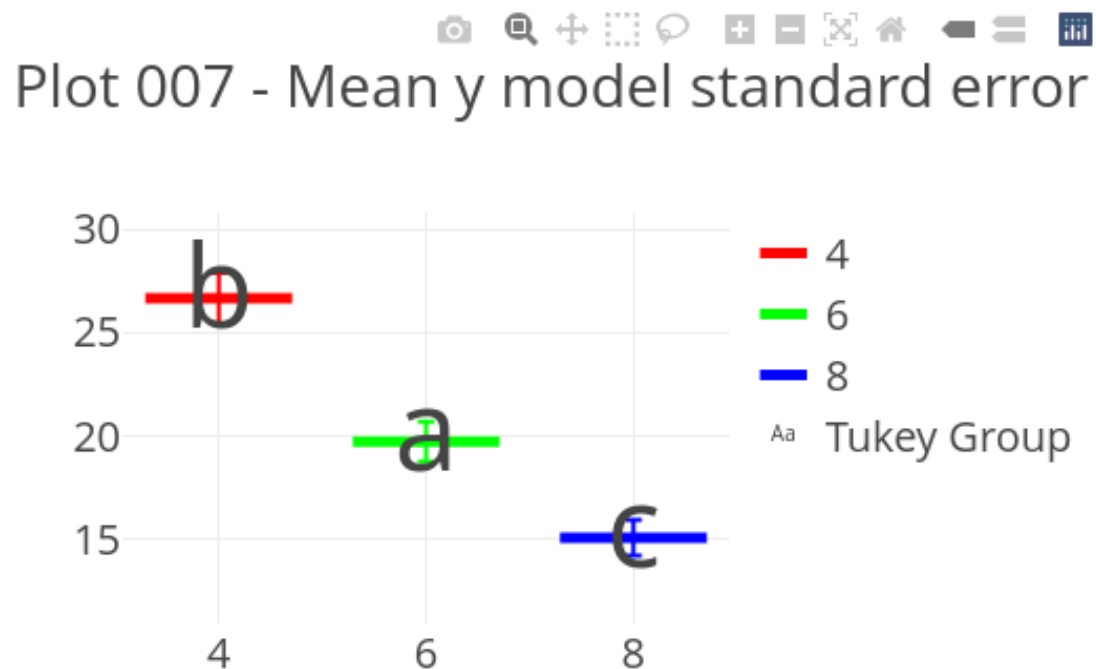
3.24.e Step 05: Interpretation of the Tukey test.

order	level	mean	group
1	4	26.66364	a
2	6	19.74286	b
3	8	15.10000	c

The means of each factor level are detailed with the standard error derived from the model. The statistical groups from the Tukey test are detailed for each factor level.

```
plot007_factor
```

```
file:///tmp/RtmpooteVR/file58cd6dcae7ce/widget58cd2a1892c0.html screenshot completed
```



3.25 FAQs

3.26 FAQs

En esta sección se dan un conjunto de preguntas tanto desde el punto de vista estadístico, informático o del diseño. Las preguntas y respuestas están mezcladas en su temática.

3.26.a 1) ¿Es mejor el test de Anova que el test t?

3.26.b 2) ¿Es lo mismo variable numérica que variable cuantitativa?

Rta: Es un error muy común. No. No es lo mismo. Una variable es numérica desde el punto de vista informático si la variable es representada con símbolos numéricos dentro del dataset. Si yo tengo una variable que tiene categorías representada con números entonces la variable no es numérica. Una variable cualitativa ordinal representada con números, como por ejemplo escala de dolor, índice de felicidad, cobertura de suelo determinada a ojo del operador, cuanto me gusta el helado de chocolate del 1 al 10, no son variable cuantitativas.

Si la variable respuesta es cualitativa nominal representada con números o cualitativa ordinal representada con números, en ambos casos la variable es numérica pero no es cuantitativa. En esos casos ingresar al test de Anova es un error grave. El test de Anova es solo para variables cuantitativas. Si se tiene un factor y una variable respuesta cualitativa nominal representada con números, debiera tal vez realizarse un test Chi Cuadrado. Si se tiene un factor y una variable respuesta cualitativa ordinal representada con números, debería utilizarse herramientas estadísticas de Distribución Libre como es el test de Kruskal-Wallis.

3) ¿Es mejor el test de Anova a 1 fActor que el Test de Kruskal-Wallis? Algo es mejor o peor en un contexto. La idea general de la bibliografía es que si hay un contexto en que podemos poner a prueba una característica de nuestra población con diferentes herramienta estadística, debemos elegir la más potente. Esto quiere decir que de las herramientas estadísticas disponibles y válidas para analizar mis datos, yo debo elegir aquella que tiene más chances de rechazar H_0 . En esta frase recalco la palabra válidas.

Si se tiene un factor con una variable respuesta, y la variable respuesta es cuantitativa, y al ingresar al test de Anova los residuos cumplen los requisitos de homogeneidad y normalidad, en ese caso yo tengo para elegir tanto Anova a 1 Factor como el Test de Kruskal-Wallis. En este contexto es mejor Anova que el test de Kruskal-Wallis.

Si se tiene un factor con una variable respuesta, y la variable respuesta es cuantitativa, y al ingresar al test de Anova los residuos no cumplen todos los requisitos de homogeneidad y normalidad, en ese caso Anova ya no es válido como opción. Mi única opción en este contexto es Kruskal-Wallis.

Si se tiene un factor con una variable respuesta, y la variable respuesta es cualitativa ordinal representada con números, Anova ya no es si quiera es una opción ya que no es válido trabajar dentro de anova con datos cualitativos (aunque estén representados con símbolos números). Mi única opción en este contexto es Kruskal-Wallis.

Si se tiene un factor con una variable respuesta, y la variable respuesta es cualitativa nominal representada con números, en este contexto ni Anova ni Kruskal-Wallis son las herramientas correctas para analizar esos datos.

3.26.c 3) ¿Es lo mismo la media de las medias que la media de toda la variable respuesta?

Rta: No. Y es también un error común. Hablemos un poco antes de la media y la mediana. Aunque la media y la mediana pueden tener el mismo valor numérico, no son lo mismo. Que ocasionalmente puedan adquirir ambas el mismo valor no las hace el mismo concepto. La media y la mediana coinciden en cualquier distribución simétrica. Cuando se tienen distribuciones simétricas como es la distribución normal, entonces coinciden media y mediana.

Algo similar ocurre con la media de las medias y la media de toda la variable respuesta. La media de las medias será igual a la media de toda la variable respuesta cuando el “n” de todos los niveles del factor sea el mismo. Solo en ese caso coinciden. En Anova tenemos μ , τ y μ_i , y entre otras cosas ocurre que la suma de los τ es cero. Eso solo ocurre si μ es estimada con la media de las medias. En Anova a 1 Factor, μ , que se le suele decir la Media General, o la Esperanza General, es la media de las medias. Puede usted fácilmente corroborar esto calculando los valores τ y la suma de los valores τ para un caso y para el otro.

4 R Script

```
---
title: "Rscience - _script_modulo"
format:
  html:
    theme:
      - custom_theme.scss
    grid:
      body-width: 2000px
      margin-width: 250px
      gutter-width: 1.5rem
    toc: true
    toc-float: true
    toc-location: right
    self-contained: true
    link-citations: true # Util para ver referencias en el margen (ver
`tuftes.html` en la galería de Quarto)

knitr:
  opts_chunk:
    collapse: false
    comment: ""
params:
  file_name: "mtcars"
  file_source: "r_source"
  var_name_rv: "mpg"
  var_name_factor: "cyl"
  alpha_value: "0.05"
  vector_ordered_levels:
    - "6"
    - "4"
    - "8"
  vector_ordered_colors:
    - "#000000"
    - "#00FF00"
    - "#0000FF"
---

# Libraries
library("htmlwidgets")
library("knitr")

# Initials
file_name <- params$"file_name"
file_source <- params$"file_source"
var_name_rv <- params$"var_name_rv"
var_name_factor <- params$"var_name_factor"
alpha_value <- as.numeric(as.character(params$"alpha_value"))
vector_ordered_levels <- params$"vector_ordered_levels"
vector_ordered_colors <- params$"vector_ordered_colors"

#vector_ordered_colors <- params$"vector_ordered_colors"
```

```

# Basics lvl 01
check_r_source      <- file_source == "r_source"
check_rscience_source <- file_source == "rscience_source"
check_excel_source  <- file_source == "excel_source"
check_csv_source     <- file_source == "csv_source"

if(check_r_source) my_dataset <- get(file_name)

#### INIT CODE ####
# # # # Section 01 - Libraries -----
library("stats")      # General Linear Models
library("agricolae")  # Tukey test
library("plotly")     # Advanced graphical functions
library("dplyr")      # Developing with %>%

# # # # Section 02 - Import dataset -----
#+++---my_dataset <- _+A+_my_import_sentence_+A+_
head(x = my_dataset, n = 5)

# # # # Section 03 - Settings -----
#+++---var_name_rv      <- "_+B+_var_name_rv_+B+_ "
#+++---var_name_factor <- "_+B+_var_name_factor_+B+_ "
#+++---alpha_value     <- _+B+_alpha_value_+B+_
#+++---vector_ordered_levels <- _+C+_vector_ordered_levels_+C+_
#+++---vector_ordered_colors <- _+C+_vector_ordered_colors_+C+_

# # # # Section 04 - Standard actions -----
# The factor must be factor data type on R.
my_dataset[,var_name_factor] <-
as.factor(as.character(my_dataset[,var_name_factor]))
my_dataset[,var_name_factor] <- factor(my_dataset[,var_name_factor], levels
= vector_ordered_levels)

# # # # Section 05 - Alpha and confidence value -----
confidence_value <- 1 - alpha_value

df_alpha_confidence <- data.frame(
  "order" = 1:2,
  "detail" = c("alpha value", "confidence value"),
  "probability" = c(alpha_value, confidence_value),
  "percentaje" = paste0(c(alpha_value, confidence_value)*100, "%")
)
df_alpha_confidence

# # # # Section 06 - Selected variables and roles -----
vector_all_var_names <- colnames(my_dataset)
vector_name_selected_vars <- c(var_name_rv, var_name_factor)
vector_rol_vars <- c("RV", "FACTOR")

```

```

df_selected_vars <- data.frame(
  "order" = 1:length(vector_name_selected_vars),
  "var_name" = vector_name_selected_vars,
  "var_number" = match(vector_name_selected_vars, vector_all_var_names),
  "var_letter" = openxlsx::int2col(match(vector_name_selected_vars,
vector_all_var_names)),
  "var_role" = vector_rol_vars,
  "doble_reference" = paste0(vector_rol_vars, "(",
vector_name_selected_vars, ")")
)
df_selected_vars

# # # Section 05 - minibase
-----
# Only selected variables.
# Only completed rows.
# Factor columns as factor object in R.
minibase <- na.omit(my_dataset[vector_name_selected_vars])
#colnames(minibase) <- vector_rol_vars
minibase[,var_name_factor] <- as.factor(minibase[,var_name_factor])
head(x = minibase, n = 5)

# # # Anova control
# 'VR' must be numeric and 'FACTOR must be factor.
df_control_minibase <- data.frame(
  "order" = 1:nrow(df_selected_vars),
  "var_name" = df_selected_vars$var_name,
  "var_role" = df_selected_vars$var_role,
  "control" = c("is.numeric()", "is.factor()"),
  "verify" = c(is.numeric(minibase[,var_name_rv]),
is.factor(minibase[,var_name_factor]))
)
df_control_minibase

# # # my_dataset and minibase reps
# Our 'n' is from minibase
df_show_n <- data.frame(
  "object" = c("my_dataset", "minibase"),
  "n_col" = c(ncol(my_dataset), ncol(minibase)),
  "n_row" = c(nrow(my_dataset), nrow(minibase))
)
df_show_n

# # # Factor info
# Default order for levels its alphabetic order.
df_factor_info <- data.frame(

```

```

"order" = 1:nlevels(minibase[,2]),
"level" = levels(minibase[,2]),
"n" = as.vector(table(minibase[,2])),
"mean" = tapply(minibase[,1], minibase[,2], mean),
"color" = vector_ordered_colors
)
rownames(df_factor_info) <- 1:nrow(df_factor_info)
df_factor_info

# # # Unbalanced reps for levels?
# Important information for Tukey.
# If reps its equal or not equal in all levels must be detailed
# on Tukey.
check_unbalanced_reps <- length(unique(df_factor_info$n)) > 1
check_unbalanced_reps

phrase_yes_unbalanced <- "The design is unbalanced in repetitions. A
correction is applied to the Tukey test."
phrase_no_unbalanced <- "The design is unbalanced in replicates. A
correction should be applied to the Tukey test."
phrase_selected_check_unbalanced <- ifelse(test = check_unbalanced_reps,
      yes = phrase_yes_unbalanced,
      no = phrase_no_unbalanced)

phrase_selected_check_unbalanced

# # # Section 06 - Anova Test
-----
# # # Anova test
the_formula <- paste0(var_name_rv, " ~ ", var_name_factor)
the_formula <- as.formula (the_formula)
lm_anova <- lm(formula = the_formula, data = minibase) #
Linear model
aov_anova <- aov(lm_anova) # R results for
anova
df_table_anova <- as.data.frame(summary(aov_anova)[[1]]) # Common anova
table
df_table_anova

# # # Standard error from model for each level
model_error_var_MSE <- df_table_anova$`Mean Sq`[2]
model_error_sd <- sqrt(model_error_var_MSE)

df_model_error <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "model_error_var_MSE" = model_error_var_MSE,

```

```

    "model_error_sd" = model_error_sd
  )
  df_model_error["model_error_se"] <- df_model_error["model_error_sd"]/
sqrt(df_model_error$n)
df_model_error

# # # Section 07 - minibase_mod
-----
# # # Detect rows on my_dataset there are on minibase
      dt_rows_my_dataset_ok <- rowSums(!
is.na(my_dataset[vector_name_selected_vars])) == ncol(minibase)

# # # Object minibase_mod and new cols
minibase_mod <- minibase
minibase_mod$"lvl_order_number" <- as.numeric(minibase_mod[,2])
      minibase_mod$"lvl_color" <-
df_factor_info$color[minibase_mod$"lvl_order_number"]
      minibase_mod$"fitted.values" <-
df_factor_info$"mean"[minibase_mod$"lvl_order_number"]
minibase_mod$"residuals" <- lm_anova$residuals
      minibase_mod$"id_my_dataset" <- c(1:nrow(my_dataset))
[dt_rows_my_dataset_ok]
minibase_mod$"id_minibase" <- 1:nrow(minibase)
minibase_mod$"studres" <- minibase_mod$"residuals"/model_error_sd

# # # Section 08 - Requeriments for
residuals-----
# # # Normality test (Shapiro-Wilk)
test_residuals_normality <- shapiro.test(minibase_mod$residuals)
test_residuals_normality

# # # Homogeinidy test (Bartlett)
the_formula_bartlett <- paste0("residuals", " ~ ", var_name_factor)
the_formula_bartlett <- as.formula(the_formula_bartlett)
test_residuals_homogeneity <- bartlett.test(the_formula_bartlett, data =
minibase_mod)
test_residuals_homogeneity

```

```

# # # Residuals variance from levels from original residuals
df_raw_error <- data.frame(
  "order" = 1:nlevels(minibase_mod[,2]),
  "level" = levels(minibase_mod[,2]),
  "n" = tapply(minibase_mod$residuals, minibase_mod[,2], length),
  "raw_error_var" = tapply(minibase_mod$residuals, minibase_mod[,2], var),
  "raw_error_sd" = tapply(minibase_mod$residuals, minibase_mod[,2], sd)
)
df_model_error["raw_error_se"] <- df_model_error["model_error_sd"]/
sqrt(df_model_error$n")
rownames(df_raw_error) <- 1:nrow(df_raw_error)
df_raw_error

phrase_info_errors <- c("Anova and Tukey use MSE from model.",
  "Bartlett use variance from raw error on each
level.",
  "Only if there is homogeneity from raw error
variances then is a good idea take desition from MSE.")
phrase_info_errors
# # # Sum for residuals
sum_residuals <- sum(minibase_mod$residuals)
sum_residuals

# # # Mean for residuals
mean_residuals <- mean(minibase_mod$residuals)
mean_residuals

#####
p_value_shapiro <- test_residuals_normality$p.value
p_value_bartlett <- test_residuals_homogeneity$p.value
p_value_anova <- df_table_anova$`Pr(>F)`[1]

vector_p_value <- c(p_value_shapiro, p_value_bartlett, p_value_anova)
vector_logic_rejected <- vector_p_value < alpha_value
vector_ho_decision <- ifelse(test = vector_logic_rejected, yes = "Ho
Rejected", "Ho no rejected")
vector_ho_rejected <- ifelse(test = vector_logic_rejected, yes = "Yes",
"No")

df_summary_anova <- data.frame(
  "test" = c("Shapiro-Wilk test", "Bartlett test", "Anova 1 way"),
  "aim" = c("Normality", "Homogeneity", "Mean"),
  "variable" = c("residuals", "residuals", var_name_rv),
  "p_value" = vector_p_value,
  "alpha_value" = c(alpha_value, alpha_value, alpha_value),
  "Decision" = vector_ho_decision
)

```



```

df_summary_anova

check_shapiro_rejected      <- p_value_shapiro < alpha_value
phrase_shapiro_yes_rejected <- "The null hypothesis of normal distribution
of residuals is rejected."
phrase_shapiro_no_rejected  <- "The null hypothesis of normal distribution
of residuals is not rejected."
phrase_shapiro_selected     <- ifelse(test = check_shapiro_rejected,
                                     yes = phrase_shapiro_yes_rejected,
                                     no  = phrase_shapiro_no_rejected)

phrase_shapiro_selected

check_bartlett_rejected     <- p_value_bartlett < alpha_value
phrase_bartlett_yes_rejected <- "The hypothesis of homogeneity of variances
(homoscedasticity) is rejected."
phrase_bartlett_no_rejected <- "The hypothesis of homogeneity of variances
(homoscedasticity) is not rejected."
phrase_bartlett_selected    <- ifelse(test = check_bartlett_rejected,
                                     yes = phrase_bartlett_yes_rejected,
                                     no  = phrase_bartlett_no_rejected)

phrase_bartlett_selected

check_ok_all_requeriments  <- sum(vector_logic_rejected[c(1,2)]) == 2
phrase_requeriments_yes_valid <- "All residual assumptions are met, so it
is valid to draw conclusions from the ANOVA test."
phrase_requeriments_no_valid  <- "Not all model assumptions are met, so it
is NOT valid to draw conclusions from the ANOVA test."
phrase_requeriments_selected  <- ifelse(test = check_ok_all_requeriments,
                                     yes = phrase_requeriments_yes_valid,
                                     no  = phrase_requeriments_no_valid)

phrase_requeriments_selected

check_anova_rejected       <- p_value_anova < alpha_value
phrase_anova_yes_rejected  <- "The null hypothesis of the ANOVA test is
rejected. There are statistically significant differences in at least one
level of the factor."
phrase_anova_no_rejected   <- "The null hypothesis of the ANOVA test is not
rejected. All levels of the factor are statistically equal."
phrase_anova_selected      <- ifelse(test = check_anova_rejected,
                                     yes = phrase_anova_yes_rejected,
                                     no  = phrase_anova_no_rejected)

phrase_anova_selected <- ifelse(
  test = check_ok_all_requeriments,
  yes = phrase_anova_selected,
  no  = "Regardless of the p-value obtained in ANOVA, it is not valid to
draw conclusions."

```

```

)

phrase_anova_selected
#####
tukey01_full_groups <- agricolae::HSD.test(y = lm_anova,
                                          trt = colnames(minibase_mod)[2],
                                          alpha = alpha_value,
                                          group = TRUE,
                                          console = FALSE,
                                          unbalanced = check_unbalanced_reps)

# # # Tukey test - Tukey pairs comparison - Full version
tukey02_full_pairs <- agricolae::HSD.test(y = lm_anova,
                                          trt = colnames(minibase_mod)[2],
                                          alpha = alpha_value,
                                          group = FALSE,
                                          console = FALSE,
                                          unbalanced = check_unbalanced_reps)

# # Original table from R about Tukey
df_tukey_original_table <- tukey01_full_groups$groups
df_tukey_original_table

# # # New table about Tukey
df_tukey_table <- data.frame(
  "level" = rownames(tukey01_full_groups$groups),
  "mean" = tukey01_full_groups$groups[,1],
  "group" = tukey01_full_groups$groups[,2]
)
df_tukey_table

# # # Section 10 - Partitioned Measures
(VR)-----
# # # Partitioned Measures of Position (VR)
df_vr_position_levels <- data.frame(
  "order" = 1:nlevels(minibase[,2]),
  "level" = levels(minibase[,2]),
  "min" = tapply(minibase[,1], minibase[,2], min),
  "mean" = tapply(minibase[,1], minibase[,2], mean),
  "Q1" = tapply(minibase[,1], minibase[,2], quantile, 0.25),
  "median" = tapply(minibase[,1], minibase[,2], median),
  "Q3" = tapply(minibase[,1], minibase[,2], quantile, 0.75),

```

```

    "max" = tapply(minibase[,1], minibase[,2], max),
    "n" = tapply(minibase[,1], minibase[,2], length)
  )

# # # Partitioned Measures of Dispersion (VR)
df_vr_dispersion_levels <- data.frame(
  "order" = 1:nlevels(minibase[,2]),
  "level" = levels(minibase[,2]),
  "range" = tapply(minibase[,1], minibase[,2], function(x){max(x) -
min(x)}),
  "variance" = tapply(minibase[,1], minibase[,2], var),
  "standard_deviation" = tapply(minibase[,1], minibase[,2], sd),
  "standard_error" = tapply(minibase[,1], minibase[,2], function(x){sd(x)/
sqrt(length(x))}),
  "n" = tapply(minibase[,1], minibase[,2], length)
)
df_vr_dispersion_levels

# # # General Measures of Position (VR)
df_vr_position_general <- data.frame(
  "min" = min(minibase[,1]),
  "mean" = mean(minibase[,1]),
  "median" = median(minibase[,1]),
  "max" = max(minibase[,1]),
  "n" = length(minibase[,1])
)
df_vr_position_general

# # # General Measures of Dispersion (VR)
df_vr_dispersion_general <- data.frame(
  "range" = max(minibase[,1]) - min(minibase[,1]),
  "variance" = var(minibase[,1]),
  "standard_deviation" = sd(minibase[,1]),
  "standard_error" = sd(minibase[,1])/(sqrt(length(minibase[,1]))),
  "n" = length(minibase[,1])
)
df_vr_dispersion_general

# # # Section 11 - Partitioned Measures
# # # (Residuals) -----
# # # Partitioned Measures of Position (residuals)
df_residuals_position_levels <- data.frame(

```

```

"order" = 1:nlevels(minibase_mod[,2]),
"level" = levels(minibase_mod[,2]),
"min" = tapply(minibase_mod$residuals, minibase_mod[,2], min),
"mean" = tapply(minibase_mod$residuals, minibase_mod[,2], mean),
"median" = tapply(minibase_mod$residuals, minibase_mod[,2], median),
"max" = tapply(minibase_mod$residuals, minibase_mod[,2], max),
"n" = tapply(minibase_mod$residuals, minibase_mod[,2], length)
)
df_residuals_position_levels

# # # Partitioned Measures of Dispersion (residuals)
df_residual_dispersion_levels <- data.frame(
  "order" = 1:nlevels(minibase_mod[,2]),
  "level" = levels(minibase_mod[,2]),
  "range" = tapply(minibase_mod$residuals, minibase_mod[,2], function(x)
{max(x) - min(x)}),
  "variance" = tapply(minibase_mod$residuals, minibase_mod[,2], var),
  "standard_deviation" = tapply(minibase_mod$residuals, minibase_mod[,2],
sd),
  "standard_error" = tapply(minibase_mod$residuals, minibase_mod[,2],
function(x){sd(x)/sqrt(length(x))}),
  "n" = tapply(minibase_mod$residuals, minibase_mod[,2], length)
)
df_residual_dispersion_levels

# # # General Measures of Position (residuals)
df_residuals_position_general <- data.frame(
  "min" = min(minibase_mod$residuals),
  "mean" = mean(minibase_mod$residuals),
  "median" = median(minibase_mod$residuals),
  "max" = max(minibase_mod$residuals),
  "n" = length(minibase_mod$residuals)
)
df_residuals_position_general

# # # General Measures of Dispersion (residuals)
df_residuals_dispersion_general <- data.frame(
  "range" = max(minibase_mod$residuals) - min(minibase_mod$residuals),
  "variance" = var(minibase_mod$residuals),
  "standard_deviation" = sd(minibase_mod$residuals),
  "standard_error" = sd(minibase_mod$residuals)/
(sqrt(length(minibase_mod$residuals))),
  "n" = length(minibase_mod$residuals)
)
df_residuals_dispersion_general

```

```
##### Means for each level
vector_est_mu_i <- df_vr_position_levels$mean
vector_est_mu_i

##### Mean of means
est_mu <- mean(vector_est_mu_i)
vector_est_mu <- rep(est_mu, length(vector_est_mu_i))
vector_est_mu

##### Tau effects
vector_est_tau_i <- vector_est_mu_i - vector_est_mu
vector_est_tau_i

##### Sum of tau effects
sum_est_tau_i <- sum(vector_est_tau_i)
sum_est_tau_i

##### Long model information on dataframe
df_anovalway_model_long <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "est_mu" = vector_est_mu,
  "est_tau_i" = vector_est_tau_i
)
df_anovalway_model_long

##### Short model information on dataframe
df_anovalway_model_short <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "est_mu_i" = vector_est_mu_i
)
df_anovalway_model_short
```

Section 13 - Special table to plots

```
# # # Table for plot001
df_table_factor_plot001 <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "mean" = tapply(minibase[,1], minibase[,2], mean),
  "min" = tapply(minibase[,1], minibase[,2], min),
  "max" = tapply(minibase[,1], minibase[,2], max),
  "sd" = tapply(minibase[,1], minibase[,2], sd),
  "var" = tapply(minibase[,1], minibase[,2], var)
)

df_table_factor_plot002 <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "mean" = tapply(minibase[,1], minibase[,2], mean),
  "model_error_sd" = df_model_error$model_error_sd
)
df_table_factor_plot002["lower_limit"] <- df_table_factor_plot002$mean -
df_table_factor_plot002$model_error_sd
df_table_factor_plot002["upper_limmit"] <- df_table_factor_plot002$mean +
df_table_factor_plot002$model_error_sd
df_table_factor_plot002["color"] <- df_factor_info$color
df_table_factor_plot002

df_table_factor_plot003 <- data.frame(
  "order" = df_factor_info$order,
  "level" = df_factor_info$level,
  "n" = df_factor_info$n,
  "mean" = tapply(minibase[,1], minibase[,2], mean),
  "model_error_se" = df_model_error$model_error_se
)
df_table_factor_plot003["lower_limit"] <- df_table_factor_plot003$mean -
df_table_factor_plot003$model_error_se
df_table_factor_plot003["upper_limmit"] <- df_table_factor_plot003$mean +
df_table_factor_plot003$model_error_se
df_table_factor_plot003["color"] <- df_factor_info$color
df_table_factor_plot003

# # # Table for plot004
```

```

df_table_factor_plot004 <- df_vr_position_levels
df_table_factor_plot004["color"] <- df_factor_info$color

# # # Table for plot005
df_table_factor_plot005 <- df_table_factor_plot004

# # # Table for plot006
df_table_factor_plot006 <- df_table_factor_plot004

df_table_factor_plot007 <- df_table_factor_plot003
correct_pos_letters <- order(df_tukey_table$level)
vector_letters <- df_tukey_table$group[correct_pos_letters]
df_table_factor_plot007["group"] <- vector_letters

# # # Table for plot006
df_table_residuals_plot001 <- data.frame(
  "order" = 1:nlevels(minibase_mod[,2]),
  "level" = levels(minibase_mod[,2]),
  "n" = tapply(minibase_mod$residuals, minibase_mod[,2], length),
  "min" = tapply(minibase_mod$residuals, minibase_mod[,2], min),
  "mean" = tapply(minibase_mod$residuals, minibase_mod[,2], mean),
  "max" = tapply(minibase_mod$residuals, minibase_mod[,2], max),
  "var" = tapply(minibase_mod$residuals, minibase_mod[,2], var),
  "sd" = tapply(minibase_mod$residuals, minibase_mod[,2], sd),
  "color" = df_factor_info$color
)
df_table_residuals_plot001

# # # Table for plot006
df_table_residuals_plot002 <- df_table_residuals_plot001

# # # Table for plot006
df_table_residuals_plot003 <- df_table_residuals_plot001

# # # Table for plot006
df_table_residuals_plot004 <- data.frame(
  "variable" = "residuals",
  "n" = length(minibase_mod$residuals),
  "min" = min(minibase_mod$residuals),
  "mean" = mean(minibase_mod$residuals),
  "max" = max(minibase_mod$residuals),
  "var" = var(minibase_mod$residuals),
  "sd" = sd(minibase_mod$residuals),
  "model_error_var_MSE" = model_error_var_MSE,
  "model_error_sd" = model_error_sd
)

phrase_coment_errors <- "Model Error (MSE) "

# # # Table for plot006
df_table_residuals_plot005 <- df_table_residuals_plot004

```

```

# # # Table for plot006
df_table_residuals_plot006 <- data.frame(
  "order" = 1:nlevels(minibase_mod[,2]),
  "level" = levels(minibase_mod[,2]),
  "n" = tapply(minibase_mod$studres, minibase_mod[,2], length),
  "min" = tapply(minibase_mod$studres, minibase_mod[,2], min),
  "mean" = tapply(minibase_mod$studres, minibase_mod[,2], mean),
  "max" = tapply(minibase_mod$studres, minibase_mod[,2], max),
  "var" = tapply(minibase_mod$studres, minibase_mod[,2], var),
  "sd" = tapply(minibase_mod$studres, minibase_mod[,2], sd),
  "color" = df_factor_info$color
)

# # # Table for plot006
df_table_residuals_plot007 <- df_table_residuals_plot006

df_table_residuals_plot008 <- data.frame(
  "variable" = "studres",
  "n" = length(minibase_mod$studres),
  "min" = min(minibase_mod$studres),
  "mean" = mean(minibase_mod$studres),
  "max" = max(minibase_mod$studres),
  "var" = var(minibase_mod$studres),
  "sd" = sd(minibase_mod$studres)
)

df_table_residuals_plot009 <- df_table_residuals_plot008

df_table_residuals_plot010 <- df_table_residuals_plot008

#####
# # # Create a new plot...
plot001_factor <- plotly::plot_ly()

# # # Plot001 - Scatter plot for VR and FACTOR on minibase_mod
*****
plot001_factor <- plotly::add_trace(p = plot001_factor,
  type = "scatter",
  mode = "markers",
  x = minibase_mod[,var_name_factor],
  y = minibase_mod[,var_name_rv],
  color = minibase_mod[,var_name_factor],
  colors = df_factor_info$color,
  marker = list(size = 15, opacity = 0.7))

# # # Title and settings...
plot001_factor <- plotly::layout(p = plot001_factor,
  title = "Plot 001 - Scatterplot",

```



```

font = list(size = 20),
margin = list(t = 100))

# # # Without zerolines
plot001_factor <- plotly::layout(p = plot001_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Plot output
plot001_factor

#####

# # # Create a new plot...
plot002_factor <- plot_ly()

# # # Adding errors...
plot002_factor <- add_trace(p = plot002_factor,
                            type = "scatter",
                            mode = "markers",
                            x = df_table_factor_plot002$level,
                            y = df_table_factor_plot002$mean,
                            color = df_table_factor_plot002$level,
                            colors = df_table_factor_plot002$color,
                            marker = list(symbol = "line-ew-open",
                                           size = 50,
                                           opacity = 1,
                                           line = list(width = 5)),
                            error_y = list(type = "data", array =
df_table_factor_plot002$model_error_sd)
)

# # # Title and settings...
plot002_factor <- plotly::layout(p = plot002_factor,
                                title = "Plot 002 - Mean and model standard
deviation",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines
plot002_factor <- plotly::layout(p = plot002_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Plot output
plot002_factor

```

```
#####

# # # Create a new plot...
plot003_factor <- plotly::plot_ly()

# # # Adding errors...
plot003_factor <- plotly::add_trace(p = plot003_factor,
                                   type = "scatter",
                                   mode = "markers",
                                   x = df_table_factor_plot003$level,
                                   y = df_table_factor_plot003$mean,
                                   color = df_table_factor_plot003$level,
                                   colors = df_table_factor_plot003$color,
                                   marker = list(symbol = "line-ew-
open",
                                                size = 50,
                                                opacity = 1,
                                                line = list(width =
5)),
                                   error_y = list(type = "data", array =
df_table_factor_plot003$model_error_se)
                                   )

# # # Title and settings...
plot003_factor <- plotly::layout(p = plot003_factor,
                                title = "Plot 003 - Mean y model standard
error",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines
plot003_factor <-plotly::layout(p = plot003_factor,
                               xaxis = list(zeroline = FALSE),
                               yaxis = list(zeroline = FALSE))

# # # Plot output
plot003_factor

#####

# # # New plotly...
plot004_factor <- plotly::plot_ly()

# # # Boxplot and info...
plot004_factor <- plotly::add_trace(p = plot004_factor,
                                   type = "box",
                                   x = df_table_factor_plot004$level ,
                                   color = df_table_factor_plot004$level,
```

```

        colors = df_table_factor_plot004$color,
        lowerfence = df_table_factor_plot004$min,
        q1 = df_table_factor_plot004$Q1,
        median = df_table_factor_plot004$median,
        q3 = df_table_factor_plot004$Q3,
        upperfence = df_table_factor_plot004$max,
        boxmean = TRUE,
        boxpoints = FALSE,
        line = list(color = "black", width =

3)
)

# # # Title and settings...
plot004_factor <- plotly::layout(p = plot004_factor,
                                title = "Plot 004 - Boxplot and means",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines...
plot004_factor <- plotly::layout(p = plot004_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Output plot004_anova...
plot004_factor

#####

all_levels <- levels(minibase_mod[,2])
n_levels <- length(all_levels)
all_color <- rainbow(length(all_levels))

plot005_factor <- plot_ly()

# Violinplot
for (k in 1:n_levels){

  # Selected values
  selected_level <- all_levels[k]
  selected_color <- all_color[k]
  dt_filas <- minibase_mod[,2] == selected_level

  # Plotting selected violinplot
  plot005_factor <- plot005_factor %>%
    add_trace(x = minibase_mod[,2][dt_filas],
              y = minibase_mod[,1][dt_filas],
              type = "violin",
              name = paste0("violin", k),
              points = "all",

```

```

        marker = list(color = selected_color),
        line = list(color = selected_color),
        fillcolor = I(selected_color)

    )

}

# Boxplot
plot005_factor <- plotly::add_trace(p = plot005_factor,
                                   type = "box",
                                   name = "boxplot",
                                   x = df_table_factor_plot005$level ,
                                   color = df_table_factor_plot005$level ,
                                   lowerfence = df_table_factor_plot005$min,
                                   q1 = df_table_factor_plot005$Q1,
                                   median = df_table_factor_plot005$median,
                                   q3 = df_table_factor_plot005$Q3,
                                   upperfence = df_table_factor_plot005$max,
                                   boxmean = TRUE,
                                   boxpoints = TRUE,
                                   fillcolor = df_table_factor_plot005$color,
                                   line = list(color = "black", width =
3),

                                   opacity = 0.5,
                                   width = 0.2)

# # # Title and settings...
plot005_factor <- plotly::layout(p = plot005_factor,
                                title = "Plot 005 - Violinplot",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines...
plot005_factor <- plotly::layout(p = plot005_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot005_factor

#####

#library(plotly)
plot006_factor <- plotly::plot_ly()

```

```

# Add traces
plot006_factor <- plotly::add_trace(p = plot006_factor,
                                   type = "violin",
                                   y = minibase_mod$VR,
                                   x = minibase_mod$FACTOR,
                                   showlegend = TRUE,
                                   side = "positive",
                                   points = "all",
                                   name = "Violinplot",
                                   color = minibase_mod$FACTOR,
                                   colors = df_table_factor_plot006$color)

# # # Title and settings...
plot006_factor <- plotly::layout(p = plot006_factor,
                                title = "Plot 006 - Scatterplot + Jitter +
Smoothed",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines...
plot006_factor <- plotly::layout(p = plot006_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot006_factor

#####

# # # Create a new plot...
plot007_factor <- plotly::plot_ly()

# # # Adding errors...
plot007_factor <- plotly::add_trace(p = plot007_factor,
                                   type = "scatter",
                                   mode = "markers",
                                   x = df_table_factor_plot007$level,
                                   y = df_table_factor_plot007$mean,
                                   color = df_table_factor_plot007$level,
                                   colors = df_table_factor_plot007$color,
                                   marker = list(symbol = "line-ew-
open",
                                                size = 50,
                                                opacity = 1,
                                                line = list(width =
5)),
                                   error_y = list(type = "data", array =

```

```

df_table_factor_plot007$model_error_se)
)

plot007_factor <- add_text(p = plot007_factor,
                           x = df_table_factor_plot007$level,
                           y = df_table_factor_plot007$mean,
                           text = df_table_factor_plot007$group, name =
"Tukey Group",
                           size = 20)

# # # Title and settings...
plot007_factor <- plotly::layout(p = plot007_factor,
                                title = "Plot 007 - Mean y model standard
error",
                                font = list(size = 20),
                                margin = list(t = 100))

# # # Without zerolines
plot007_factor <- plotly::layout(p = plot007_factor,
                                xaxis = list(zeroline = FALSE),
                                yaxis = list(zeroline = FALSE))

# # # Plot output
plot007_factor

#####- - -

##### DESDE AAAAAAAAAAAAAAAAAAAAAA
# # # Create a new plot...
plot001_residuals <- plotly::plot_ly()

# # # Plot001 - Scatter plot for VR and FACTOR on minibase_mod
*****
plot001_residuals <- plotly::add_trace(p = plot001_residuals,
                                     type = "scatter",
                                     mode = "markers",
                                     x = minibase_mod$FACTOR,
                                     y = minibase_mod$residuals,
                                     color = minibase_mod$FACTOR,
                                     colors = df_factor_info$color,
                                     marker = list(size = 15, opacity =
0.7))

# # # Title and settings...
plot001_residuals <- plotly::layout(p = plot001_residuals,
                                    title = "Plot 001 - Scatterplot -
Residuals",
                                    font = list(size = 20),
                                    margin = list(t = 100))

```

```

# # # Without zerolines
plot001_residuals <- plotly::layout(p = plot001_residuals,
                                   xaxis = list(zeroline = FALSE),
                                   yaxis = list(zeroline = TRUE))

# # # Plot output
plot001_residuals

#library(plotly)
plot002_residuals <- plotly::plot_ly()

# Add traces
plot002_residuals <- plotly::add_trace(p = plot002_residuals,
                                       type = "violin",
                                       y = minibase_mod$residuals,
                                       x = minibase_mod$FACTOR,
                                       showlegend = TRUE,
                                       side = "positive",
                                       points = "all",
                                       name = "Violinplot",
                                       color = minibase_mod$FACTOR,
                                       colors = df_table_residuals_plot002$color)

# # # Title and settings...
plot002_residuals <- plotly::layout(p = plot002_residuals,
                                   title = "Plot 002 - Residuals -
Scatterplot + Jitter + Smoothed",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# # # Without zerolines...
plot002_residuals <- plotly::layout(p = plot002_residuals,
                                   xaxis = list(zeroline = FALSE),
                                   yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot002_residuals

```

```

plot003_residuals <- plotly::plot_ly()

# Add traces
plot003_residuals <- plotly::add_trace(p = plot003_residuals,
                                     type = "violin",
                                     x = minibase_mod$residuals,
                                     showlegend = TRUE,
                                     side = "positive",
                                     points = FALSE,
                                     #name = levels(minibase_mod$FACTOR)
[minibase_mod$lvl_order_number],
                                     color = minibase_mod$FACTOR,
                                     colors = df_table_residuals_plot003$color)

# # # Title and settings...
plot003_residuals <- plotly::layout(p = plot003_residuals,
                                   title = "Plot 003 - Residuals -
Scatterplot + Jitter + Smoothed",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# # # Without zerolines...
plot003residuals <- plotly::layout(p = plot003_residuals,
                                   xaxis = list(zeroline = FALSE),
                                   yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot003_residuals

plot004_residuals <- plotly::plot_ly()

# Add traces
plot004_residuals <- plotly::add_trace(p = plot004_residuals,
                                     type = "violin",
                                     x = minibase_mod$residuals,
                                     #x = minibase_mod$FACTOR,
                                     showlegend = TRUE,
                                     side = "positive",
                                     points = "all",
                                     name = " ")#

#color = minibase_mod$FACTOR,
#colors = df_table_factor_plot006$color)

```



```

# # # Title and settings...
plot004_residuals <- plotly::layout(p = plot004_residuals,
                                  title = "Plot 004 - Residuals",
                                  font = list(size = 20),
                                  margin = list(t = 100))

# # # Without zerolines...
plot004_residuals <- plotly::layout(p = plot004_residuals,
                                  xaxis = list(zeroline = TRUE),
                                  yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot004_residuals

# - el 5
qq_info <- EnvStats::qqPlot(x = minibase_mod$residuals, plot.it = F,
                           param.list = list(mean = mean(minibase_mod$residuals),
                                             sd = sd(minibase_mod$residuals)))

cuantiles_teoricos <- qq_info$x
cuantiles_observados <- qq_info$y

#library(plotly)
plot005_residuals <- plotly::plot_ly()

# Crear el gráfico QQ plot
plot005_residuals <- add_trace(p = plot005_residuals,
                              x = cuantiles_teoricos,
                              y = cuantiles_observados,
                              type = 'scatter', mode = 'markers',
                              marker = list(color = 'blue'),
                              name = "points")

# Agregar la línea de identidad
pendiente <- 1
intercepto <- 0

# Calcular las coordenadas de los extremos de la línea de identidad
x_extremos <- range(cuantiles_teoricos)
y_extremos <- pendiente * x_extremos + intercepto

# Agregar la recta de identidad
plot005_residuals <- add_trace(p = plot005_residuals,
                              x = x_extremos,
                              y = y_extremos,
                              type = 'scatter',
                              mode = 'lines',

```

```

        line = list(color = 'red'),
        name = "identity")

# Establecer etiquetas de los ejes
# plot007_residuals <- layout(p = plot007_residuals,
#                               axis = list(title = 'Expected quantiles'),
#                               axis = list(title = 'Observed quantiles'))

plot005_residuals <- plotly::layout(p = plot005_residuals,
                                   title = "Plot 005 - QQ Plot Residuals",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# Mostrar el gráfico
plot005_residuals
# - Fin el 5

# # # Create a new plot...
plot006_residuals <- plotly::plot_ly()

# # # Plot001 - Scatter plot for VR and FACTOR on minibase_mod
*****
plot006_residuals <- plotly::add_trace(p = plot006_residuals,
                                     type = "scatter",
                                     mode = "markers",
                                     x = minibase_mod$fitted.values,
                                     y = minibase_mod$residuals,
                                     color = minibase_mod$FACTOR,
                                     colors = df_factor_info$color,
                                     marker = list(size = 15, opacity =
0.7))

# # # Title and settings...
plot006_residuals <- plotly::layout(p = plot006_residuals,
                                   title = "Plot 006 - Scatterplot -
Residuals vs Fitted.values",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# # # Without zerolines
plot006_residuals <- plotly::layout(p = plot006_residuals,
                                   axis = list(zeroline = FALSE),
                                   axis = list(zeroline = TRUE))

# # # Plot output
plot006_residuals

```

```

# # # Create a new plot...
plot007_residuals <- plotly::plot_ly()

# # # Plot001 - Scatter plot for VR and FACTOR on minibase_mod
*****
plot007_residuals <- plotly::add_trace(p = plot007_residuals,
                                     type = "scatter",
                                     mode = "markers",
                                     x = minibase_mod$FACTOR,
                                     y = minibase_mod$studres,
                                     color = minibase_mod$FACTOR,
                                     colors = df_factor_info$color,
                                     marker = list(size = 15, opacity =
0.7))

# # # Title and settings...
plot007_residuals <- plotly::layout(p = plot007_residuals,
                                   title = "Plot 007 - Scatterplot -
Studentized Residuals",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# # # Without zerolines
plot007_residuals <- plotly::layout(p = plot007_residuals,
                                   xaxis = list(zeroline = FALSE),
                                   yaxis = list(zeroline = TRUE))

# # # Plot output
plot007_residuals

#library(plotly)
plot008_residuals <- plotly::plot_ly()

# Add traces
plot008_residuals <- plotly::add_trace(p = plot008_residuals,
                                     type = "violin",
                                     x = minibase_mod$studres,
                                     #x = minibase_mod$FACTOR,
                                     showlegend = TRUE,
                                     side = "positive",
                                     points = "all",

```

```

name = " ")#

#color = minibase_mod$FACTOR,
#colors = df_table_factor_plot006$color)

# # # Title and settings...
plot008_residuals <- plotly::layout(p = plot008_residuals,
                                   title = "Plot 008 - Studentized
Residuals",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# # # Without zerolines...
plot008_residuals <- plotly::layout(p = plot008_residuals,
                                   xaxis = list(zeroline = TRUE),
                                   yaxis = list(zeroline = FALSE))

# # # Output plot003_anova...
plot008_residuals

# el 9

x <- seq(-4, 4, length.out = 100)
y <- dnorm(x, mean = 0, 1)
# x <- x*model_error_sd
densidad_suavizada <- density(x, kernel = "gaussian", adjust = 0.5)
hist_data_studres <- hist(minibase_mod$studres, plot = FALSE)
hist_data_studres$"rel_freq" <- hist_data_studres$counts/
sum(hist_data_studres$counts)

densidad_studres <- density(x = minibase_mod$studres, kernel = "gaussian",
adjust = 0.5)

#library(plotly)
plot009_residuals <- plotly::plot_ly()

# plot005_residuals <- add_trace(p = plot005_residuals,
#                               x = densidad_studres$x,
#                               y = densidad_studres$y,
#                               type = 'scatter',
#                               mode = 'lines',
#                               name = 'densidad_studres')

plot009_residuals <- add_trace(p = plot009_residuals,
                              x = x,
                              y = y,

```

```

        type = 'scatter',
        mode = 'lines',
        name = 'Normal Standard')

# # Add traces
# plot005_residuals <- plotly::add_trace(p = plot005_residuals,
#                                       type = "violin",
#                                       x = minibase_mod$residuals,
#                                       #x = minibase_mod$FACTOR,
#                                       showlegend = TRUE,
#                                       side = "positive",
#                                       points = FALSE,
#                                       name = "violinplot")#

plot009_residuals <- plotly::add_trace(p = plot009_residuals,
                                       type = "bar",
                                       x = hist_data_studres$"mids",
                                       y = hist_data_studres$"density",
                                       name = "hist - studres")

plot009_residuals <- plotly::layout(p = plot009_residuals,
                                   bargap = 0)

plot009_residuals <- plotly::layout(p = plot009_residuals,
                                   title = "Plot 009 - Studres Distribution",
                                   font = list(size = 20),
                                   margin = list(t = 100))

plot009_residuals
# fin el 9

qq_info <- EnvStats::qqPlot(x = minibase_mod$studres, plot.it = F,
                            param.list = list(mean = 0,
                                                sd = 1))

cuantiles_teoricos <- qq_info$x
cuantiles_observados <- qq_info$y

#library(plotly)
plot010_residuals <- plotly::plot_ly()

# Crear el gráfico QQ plot
plot010_residuals <- add_trace(p = plot010_residuals,
                              x = cuantiles_teoricos,
                              y = cuantiles_observados,

```

```

                                type = 'scatter', mode = 'markers',
                                marker = list(color = 'blue'),
                                name = "points")

# Agregar la línea de identidad
pendiente <- 1
intercepto <- 0

# Calcular las coordenadas de los extremos de la línea de identidad
x_extremos <- range(cuantiles_teoricos)
y_extremos <- pendiente * x_extremos + intercepto

# Agregar la recta de identidad
plot010_residuals <- add_trace(p = plot010_residuals,
                              x = x_extremos,
                              y = y_extremos,
                              type = 'scatter',
                              mode = 'lines',
                              line = list(color = 'red'),
                              name = "identity")

# Establecer etiquetas de los ejes
# plot007_residuals <- layout(p = plot007_residuals,
#                             #
#                             xaxis = list(title = 'Expected quantiles'),
#                             #
#                             yaxis = list(title = 'Observed quantiles'))

plot010_residuals <- plotly::layout(p = plot010_residuals,
                                   title = "Plot 010 - QQ Plot - studres",
                                   font = list(size = 20),
                                   margin = list(t = 100))

# Mostrar el gráfico
plot010_residuals

```

5 Others

5.1 Stock

El script realiza todo lo relacionado al análisis de datos enmarcado en un diseño a un factor de efecto fijos completamente aleatorizado.

El script realiza:

- 4 test estadísticos.
- 15 gráficos.
- 20 tablas.

Cuando se habla de realizar el un Anova, en realidad no es solo realizar el Test de Análisis de la varianza, sino el test en si mismo con todos los anexos y accesorios indispensables.

Un poco más en detalle, el listado es el siguiente:

- Test ANOVA a 1 Factor.
- Un test de comparación múltiple (Test de Tukey).
- Test de Normalidad Shapiro-Wilks para los residuos.
- Test de Homogeneidad de Varianzas de Bartlett para los residuos.
- Estadísticas descriptivas para una variable cuantitativa particionada por lo niveles del factor.
- Estadísticas descriptivas aplicados a los residuos en su totalidad y aplicados a los residuos particionados por el factor.

Rscience aborda toda la complejidad relacionada a un análisis de la varianza a 1 factor ya sea dentro del test Anova, su test de comparaciones múltiple, los test estadísticos para sus requisitos y para la generación de todas las tablas y gráficos asociados.

El script realiza un manejo explícito de los datos faltantes, mediante la utilización de la función `na.omit()` previo al ingreso de los datos a las funciones estadísticas. De esa manera se maneja todo el tiempo solo a las filas de datos que efectivamente ingresan al test estadístico, a tablas y a gráficos.

El script contiene todos los criterios estadísticos relacionados a la toma de decisiones de para ANova a 1 Factor, otorgando al usuario frases directas y explícitas sobre cómo interpretar sus datos.

Rscience es tajante, directo y claro en cuanto a la toma de decisiones. Deben tomarse las frases otorgadas como frases que orientan en una buena dirección y sentido a un operador. Rscience solo facilita interpretaciones estadísticas, luego el operador deberá terminar de entender la situación bajo análisis para poder dar interpretaciones de esos resultados en su marco de trabajo o estudio.

Algunos detalles particularmente importantes del script 003 para ANova a 1 Factor, Efectos Fijos, Modelo Lineal General, son los gráficos generados para los niveles del factor utilizando la media de cada nivel y el error estándar obtenido del modelo y no del error estándar obtenido de los datos. La bibliografía estadística indica que esa es la forma correcta de plotear gráficamente lo que tanto anova como el test de comparaciones están realizando analíticamente. Sin embargo en ninguno de los softwares clásicos de estadística ha sido posible nunca obtener dicho gráfico.

También es un gran alivio para el operador poder elegir el orden de presentación de los niveles del factor, ya que por defecto todos los softwares realizan las tablas y los gráficos con el orden alfabético sin la posibilidad de poder realizar este tipo de cambios.

5.2 Software

The fundamental pillars of Rscience are **Reproducibility**, **Transparency**, and **Open Access**. We strongly believe that adherence to these principles helps make science better, more reliable, and more objective.

This section provides a complete digital footprint of the computational environment used to generate this report.

By documenting the exact R version, Rscience version, the operating system configuration, and the precise list of installed packages, we guarantee maximum transparency and portability of the analysis, ensuring that the results presented can be accurately validated and replicated by any third party.

To achieve true, byte-for-byte reproducibility, a set of explicit commands leveraging the rix library is provided. These statements enable users to install the exact version of R and Rscience used when generating this report, along with every required dependency package locked to its specific version. This process eliminates environmental variance and ensures the analysis can be faithfully recreated across different systems.

5.2.a R - Session Info

Shows the currently loaded packages and R environment configuration for this specific analysis. This includes R version, locale settings, and attached packages that are actively being used in this report.

```
sessionInfo()
```

```
R version 4.4.3 (2025-02-28)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 22.04.5 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=it_IT.UTF-8       LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=it_IT.UTF-8   LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=it_IT.UTF-8      LC_NAME=C
 [9] LC_ADDRESS=C              LC_TELEPHONE=C
[11] LC_MEASUREMENT=it_IT.UTF-8 LC_IDENTIFICATION=C

time zone: Europe/Rome
tzcode source: system (glibc)

attached base packages:
```



```

[1] stats      graphics  grDevices utils      datasets  methods   base

other attached packages:
[1] rlang_1.1.6      dplyr_1.1.4      plotly_4.11.0     ggplot2_4.0.0
[5] agricolae_1.3-7  htmlwidgets_1.6.4 DT_0.34.0          htmltools_0.5.8.1
[9] fontawesome_0.5.3 webshot2_0.1.2    kableExtra_1.4.0  knitr_1.50

loaded via a namespace (and not attached):
[1] tidyr_1.3.1      generics_0.1.4    xml2_1.4.0         EnvStats_3.1.0
[5] stringi_1.8.7    lattice_0.22-5    digest_0.6.37      magrittr_2.0.3
[9] evaluate_1.0.3   grid_4.4.3        RColorBrewer_1.1-3 fastmap_1.2.0
[13] jsonlite_2.0.0   zip_2.3.3         processx_3.8.6     chromote_0.5.1
[17] ps_1.9.1         promises_1.3.3    httr_1.4.7         purrr_1.1.0
[21] crosstalk_1.2.2  viridisLite_0.4.2 scales_1.4.0       lazyeval_0.2.2
[25] textshaping_1.0.4 cli_3.6.5         withr_3.0.2        yaml_2.3.10
[29] tools_4.4.3      AlgDesign_1.2.1.2 vctrs_0.6.5        R6_2.6.1
[33] lifecycle_1.0.4  stringr_1.5.2     MASS_7.3-65        cluster_2.1.8.1
[37] pkgconfig_2.0.3  openxlsx_4.2.8    pillar_1.11.1      later_1.4.4
[41] gtable_0.3.6     data.table_1.17.8 glue_1.8.0          Rcpp_1.1.0
[45] systemfonts_1.3.1 xfun_0.53         tibble_3.3.0       tidyselect_1.2.1
[49] rstudioapi_0.17.1 farver_2.1.2      nlme_3.1-168       websocket_1.4.4
[53] rmarkdown_2.30   svglite_2.2.1     compiler_4.4.3     S7_0.2.0

```

5.2.b R Version

R.version

Parameters	values
platform	x86_64-pc-linux-gnu
arch	x86_64
os	linux-gnu
system	x86_64, linux-gnu
status	
major	4
minor	4.3
year	2025
month	02
day	28
svn rev	87843
language	R
version.string	R version 4.4.3 (2025-02-28)
nickname	Trophy Case

5.2.c R - Installed Libraries

Lists all packages installed in the R environment, providing a complete inventory of available tools. This comprehensive list helps recreate the computational environment for reproducibility purposes.

```
installed.packages()
```

```
file:///tmp/RtmpooteVR/file58cd77cdd8ab/widget58cd16405fa9.html screenshot  
completed
```

Show 10 entries

Search:

	Package	LibPath
	agricolae	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	AlgDesign	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	AsioHeaders	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	askpass	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	base64enc	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	bit	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	bit64	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	brew	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	brio	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu
	bslib	/home/david/.cache/R/renv/library/exp0192fcf2f8/linux-ubuntu-jammy/R-4.4/x86_64-linux-gnu

Showing 1 to 10 of 185 entries

Previous 1 2 3 4 5 ... 19
Next

5.3 Bibliography

Algo hay que colocar...

5.4 Cita

Algo hay que colocar...

6 Rscience Code

```
# Libraries
library("htmlwidgets")
library("knitr")

# Initials
file_name <- params$"file_name"
file_source <- params$"file_source"
var_name_rv <- params$"var_name_rv"
var_name_factor <- params$"var_name_factor"
alpha_value <- as.numeric(as.character(params$"alpha_value"))
vector_ordered_levels <- params$"vector_ordered_levels"
vector_ordered_colors <- params$"vector_ordered_colors"

#vector_ordered_colors <- params$"vector_ordered_colors"

# Basics lvl 01
check_r_source <- file_source == "r_source"
check_rscience_source <- file_source == "rscience_source"
check_excel_source <- file_source == "excel_source"
check_csv_source <- file_source == "csv_source"

if(check_r_source) my_dataset <- get(file_name)
```

```
#### INIT CODE ####
# # # # Section 01 - Libraries -----
library("stats") # General Linear Models
library("agricolae") # Tukey test
library("plotly") # Advanced graphical functions
library("dplyr") # Developing with %>%
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2

order	detail	probability	percentaje
1	1 alpha value	0.05	5%
2	2 confidence value	0.95	95%

	order	var_name	var_number	var_letter	var_role	doble_reference
1	1	mpg	1	A	RV	RV(mpg)
2	2	cyl	2	B	FACTOR	FACTOR(cyl)

		mpg	cyl
Mazda	RX4	21.0	6
Mazda	RX4 Wag	21.0	6
Datsun	710	22.8	4
Hornet	4 Drive	21.4	6
Hornet	Sportabout	18.7	8

	order	var_name	var_role	control	verify
1	1	mpg	RV	is.numeric()	TRUE
2	2	cyl	FACTOR	is.factor()	TRUE

	object	n_col	n_row
1	my_dataset	11	32
2	minibase	2	32

	order	level	n	mean	color
1	1	6	7	19.74286	#FF0000
2	2	4	11	26.66364	#00FF00
3	3	8	14	15.10000	#0000FF

[1] TRUE

[1] "The design is unbalanced in repetitions. A correction is applied to the Tukey test."

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cyl	2	824.7846	412.39230	39.69752	4.978919e-09
Residuals	29	301.2626	10.38837	NA	NA

	Sources of Variation	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cyl	Factor	2	824.7846	412.39230	39.69752	4.978919e-09
Residuals	Error	29	301.2626	10.38837	NA	NA
Total	Total	31	1126.0472	36.32410	NA	NA

	order	level	n	model_error_var_MSE	model_error_sd	model_error_se
1	1	6	7	10.38837	3.223099	1.2182168
2	2	4	11	10.38837	3.223099	0.9718008
3	3	8	14	10.38837	3.223099	0.8614094

Shapiro-Wilk normality test

```
data: minibase_mod$residuals
W = 0.97065, p-value = 0.5177
```

Bartlett test of homogeneity of variances

```
data: residuals by cyl
Bartlett's K-squared = 8.3934, df = 2, p-value = 0.01505
```

	order	level	n	raw_error_var	raw_error_sd
1	1	6	7	2.112857	1.453567
2	2	4	11	20.338545	4.509828
3	3	8	14	6.553846	2.560048

```
[1] "Anova and Tukey use MSE from model."
[2] "Bartlett use variance from raw error on each level."
[3] "Only if there is homogeneity from raw error variances then is a good
idea take desition from MSE."
```

```
[1] -4.191092e-15
```

```
[1] -1.309716e-16
```

	test	aim	variable	p_value	alpha_value
1	Shapiro-Wilk test	Normality	residuals	5.176650e-01	0.05
2	Bartlett test	Homogeneity	residuals	1.504518e-02	0.05
3	Anova 1 way	Mean	mpg	4.978919e-09	0.05
	Decision				
1	Ho no rejected				
2	Ho Rejected				
3	Ho Rejected				

```
[1] "The null hypothesis of normal distribution of residuals is not rejected."
```

```
[1] "The hypothesis of homogeneity of variances (homoscedasticity) is
rejected."
```

```
[1] "Not all model assumptions are met, so it is NOT valid to draw conclusions
from the ANOVA test."
```

[1] "Regardless of the p-value obtained in ANOVA, it is not valid to draw conclusions."

```
      mpg groups
4 26.66364      a
6 19.74286      b
8 15.10000      c
```

```
order level    mean group
1      1      4 26.66364    a
2      2      6 19.74286    b
3      3      8 15.10000    c
```

```
order level min    mean    Q1 median    Q3 max  n
6      1      6 17.8 19.74286 18.65   19.7 21.00 21.4  7
4      2      4 21.4 26.66364 22.80   26.0 30.40 33.9 11
8      3      8 10.4 15.10000 14.40   15.2 16.25 19.2 14
```

```
order level range  variance standard_deviation standard_error  n
6      1      6   3.6   2.112857             1.453567         0.5493967  7
4      2      4  12.5  20.338545             4.509828         1.3597642 11
8      3      8   8.8   6.553846             2.560048         0.6842016 14
```

```
min    mean median max  n
1 10.4 20.09062  19.2 33.9 32
```

```
range variance standard_deviation standard_error  n
1 23.5 36.3241             6.026948         1.065424 32
```

```
order level    min    mean    median    max  n
6      1      6 -1.942857 3.410101e-16 -0.04285714 1.657143  7
4      2      4 -5.263636 -1.918446e-16 -0.66363636 7.236364 11
8      3      8 -4.700000 -3.191891e-16  0.10000000 4.100000 14
```

```
order level range  variance standard_deviation standard_error  n
6      1      6   3.6   2.112857             1.453567         0.5493967  7
4      2      4  12.5  20.338545             4.509828         1.3597642 11
8      3      8   8.8   6.553846             2.560048         0.6842016 14
```

```
min    mean    median    max  n
1 -5.263636 -1.309716e-16 0.02857143 7.236364 32
```

	range	variance	standard_deviation	standard_error	n
1	12.5	9.718148	3.117394	0.5510827	32

```
[1] 19.74286 26.66364 15.10000
```

```
[1] 20.50216 20.50216 20.50216
```

```
[1] -0.7593074 6.1614719 -5.4021645
```

```
[1] -5.329071e-15
```

	order	level	n	est_mu	est_tau_i
1	1	6	7	20.50216	-0.7593074
2	2	4	11	20.50216	6.1614719
3	3	8	14	20.50216	-5.4021645

	order	level	n	est_mu_i
1	1	6	7	19.74286
2	2	4	11	26.66364
3	3	8	14	15.10000

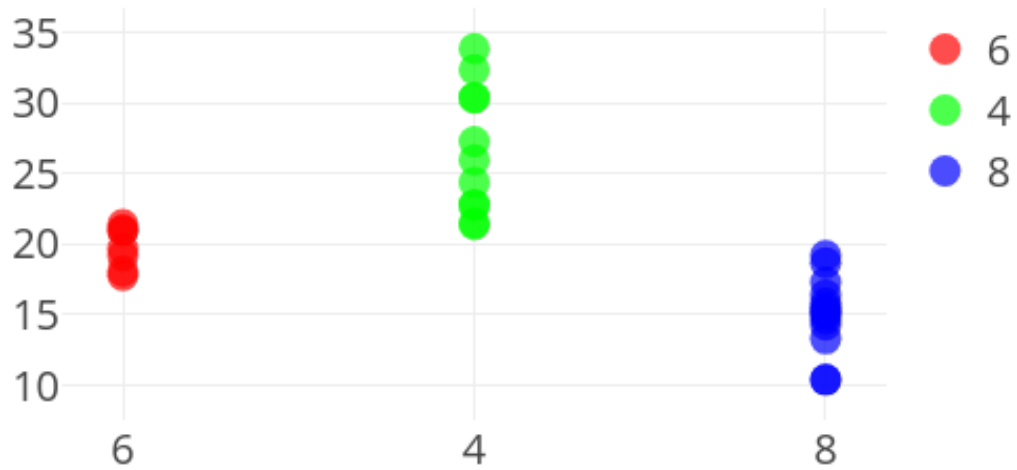
	order	level	n	mean	model_error_sd	lower_limit	upper_limit	color
6	1	6	7	19.74286	3.223099	16.51976	22.96596	#FF0000
4	2	4	11	26.66364	3.223099	23.44054	29.88674	#00FF00
8	3	8	14	15.10000	3.223099	11.87690	18.32310	#0000FF

	order	level	n	mean	model_error_se	lower_limit	upper_limit	color
6	1	6	7	19.74286	1.2182168	18.52464	20.96107	#FF0000
4	2	4	11	26.66364	0.9718008	25.69184	27.63544	#00FF00
8	3	8	14	15.10000	0.8614094	14.23859	15.96141	#0000FF

	order	level	n	min	mean	max	var	sd	color
6	1	6	7	-1.942857	3.410101e-16	1.657143	2.112857	1.453567	#FF0000
4	2	4	11	-5.263636	-1.918446e-16	7.236364	20.338545	4.509828	#00FF00
8	3	8	14	-4.700000	-3.191891e-16	4.100000	6.553846	2.560048	#0000FF

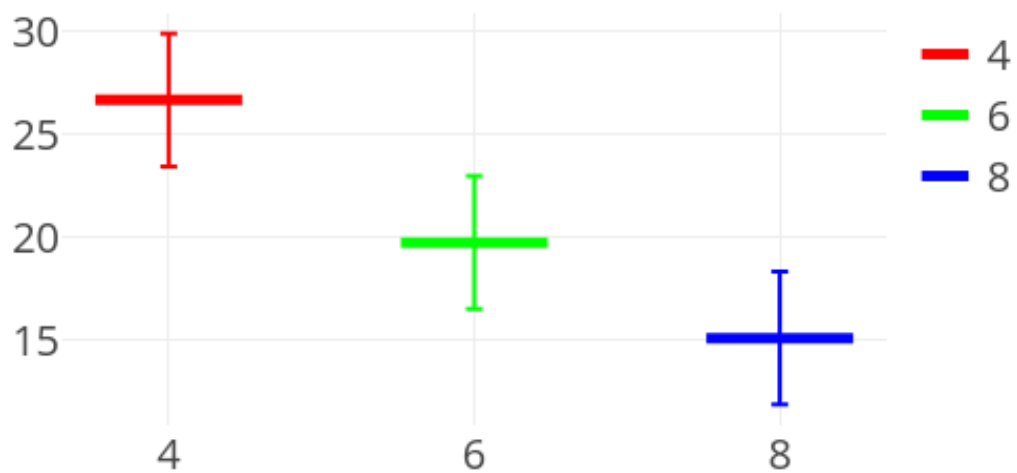
```
file:///tmp/RtmpooteVR/file58cd7b415894/widget58cd4f7682d.html screenshot completed
```


Plot 001 - Scatterplot



file:///tmp/RtmpooteVR/file58cd1d0f184d/widget58cd1ee8684d.html screenshot completed

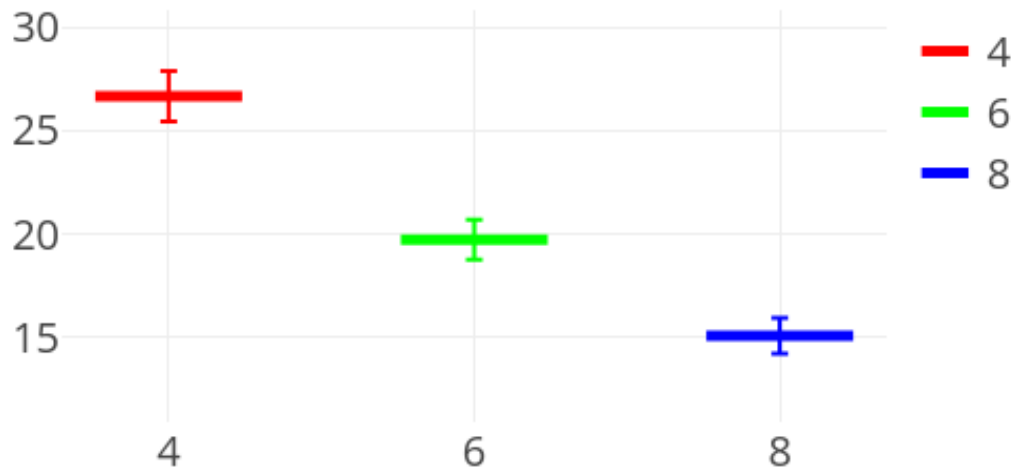
Plot 002 - Mean and model standard deviation



file:///tmp/RtmpooteVR/file58cd57f33f79/widget58cd7963a66d.html screenshot completed



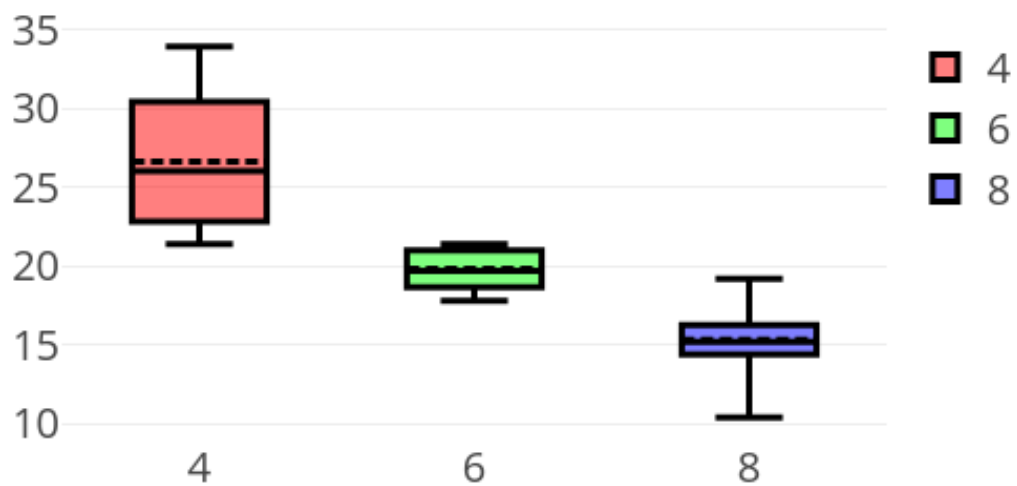
Plot 003 - Mean y model standard error



file:///tmp/RtmpooteVR/file58cd4671a47c/widget58cd5cbaf233.html screenshot completed



Plot 004 - Boxplot and means



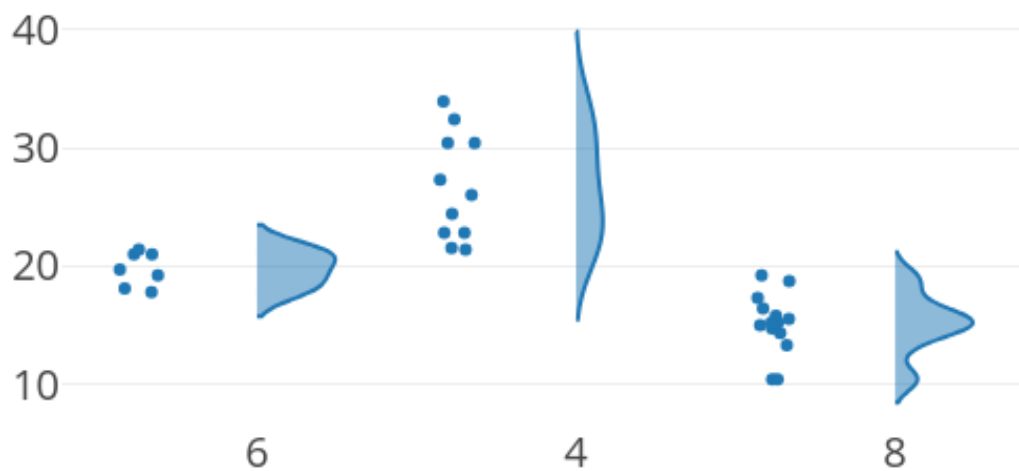
file:///tmp/RtmpooteVR/file58cd7252292d/widget58cd18b89cad.html screenshot completed

Plot 005 - Violinplot



file:///tmp/RtmpooteVR/file58cd3c95c581/widget58cd1cd3be48.html screenshot completed

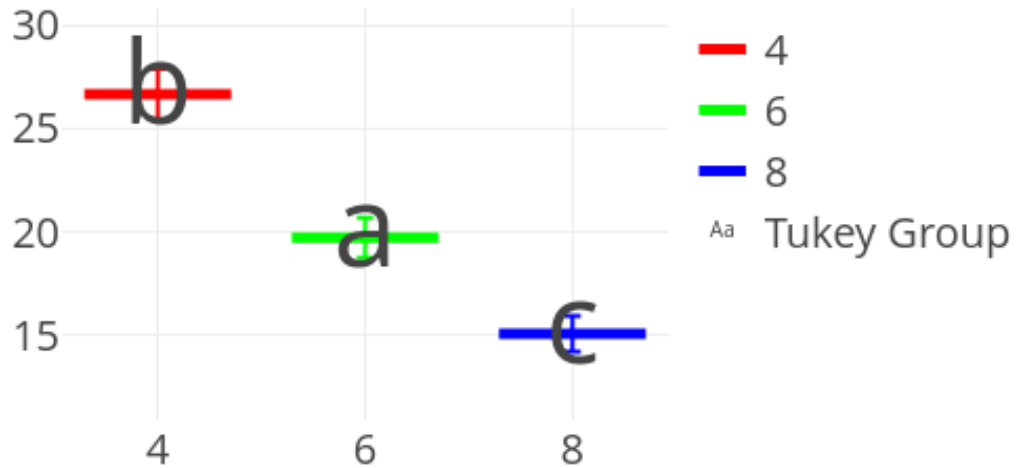
Plot 006 - Scatterplot + Jitter + Smoothed



file:///tmp/RtmpooteVR/file58cd4afefd8a/widget58cd327aae0a.html screenshot completed



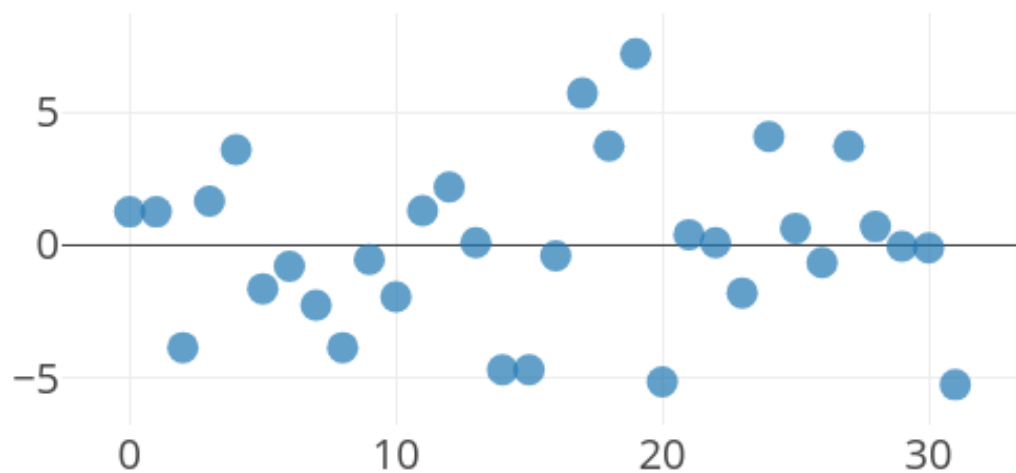
Plot 007 - Mean y model standard error



file:///tmp/RtmpooteVR/file58cd76c8b846/widget58cd5c4a9fb3.html screenshot completed



Plot 001 - Scatterplot - Residuals



file:///tmp/RtmpooteVR/file58cd614207e0/widget58cd7bcf4390.html screenshot completed



02 - Residuals - Scatterplot + Jitter + Smo

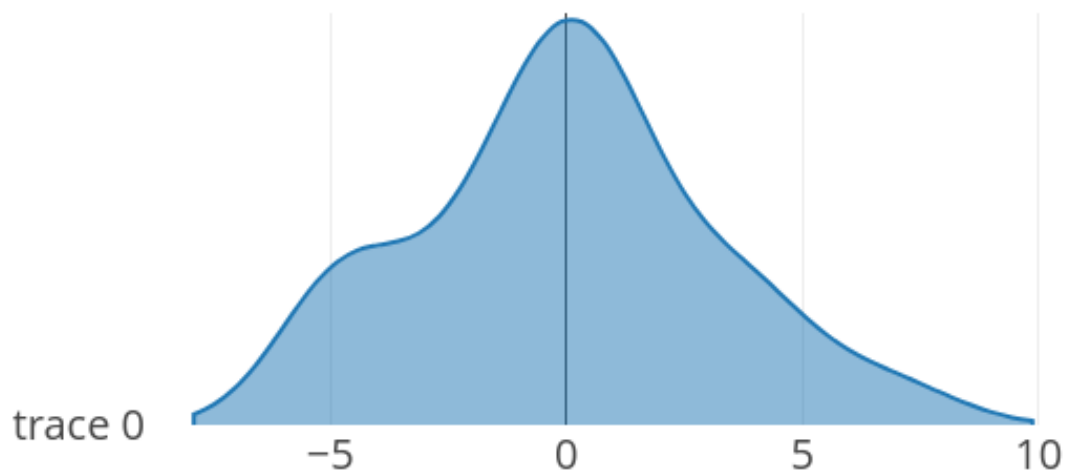


Violinplot

file:///tmp/RtmpooteVR/file58cd1ab7abdd/widget58cda786efe.html screenshot completed



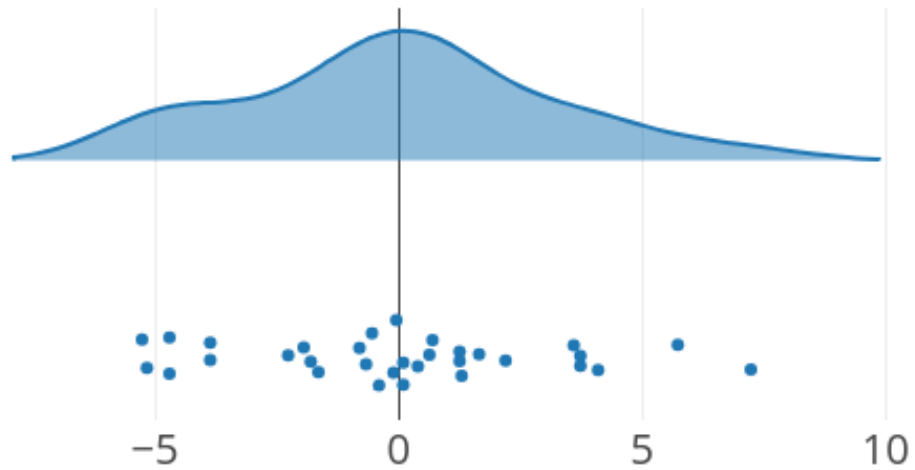
03 - Residuals - Scatterplot + Jitter + Smo



file:///tmp/RtmpooteVR/file58cd3dc156c/widget58cd5ea23eab.html screenshot completed



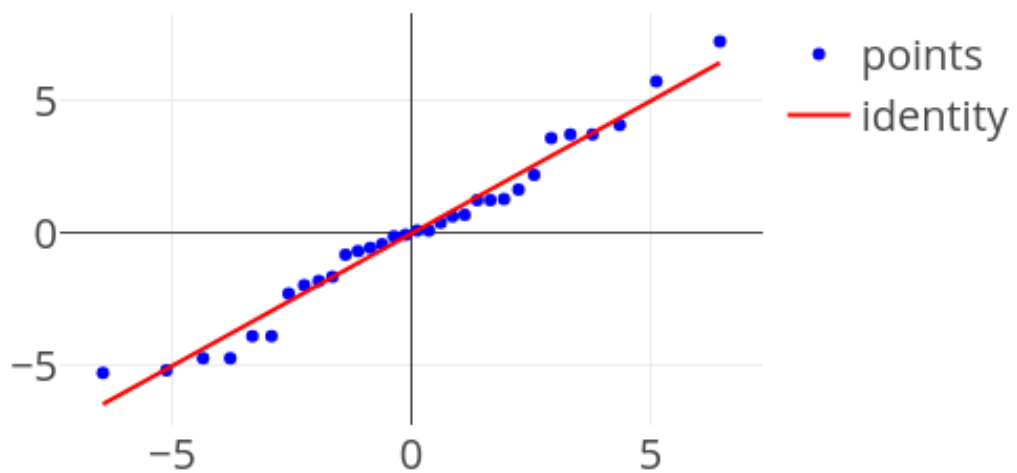
Plot 004 - Residuals



file:///tmp/RtmpooteVR/file58cd3b5d30de/widget58cd4fac9422.html screenshot completed



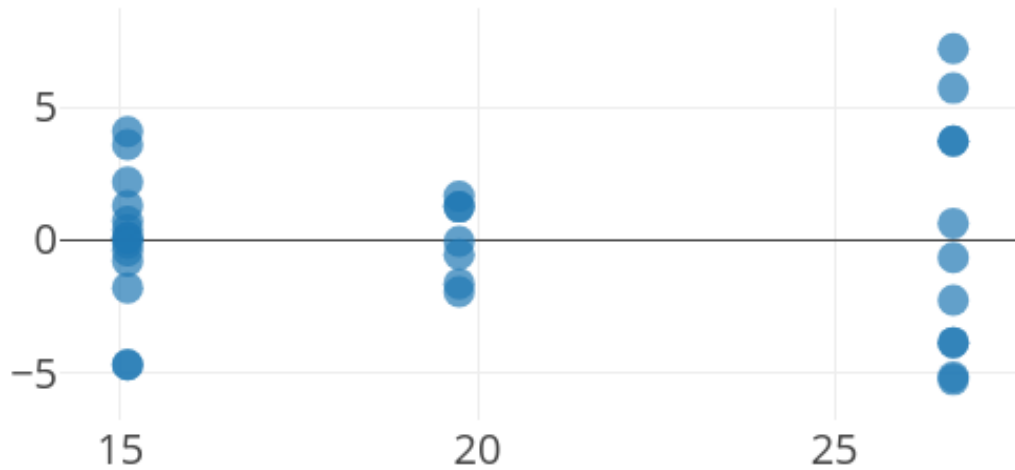
Plot 005 - QQ Plot Residuals



file:///tmp/RtmpooteVR/file58cd686530cf/widget58cd4c1845b4.html screenshot completed



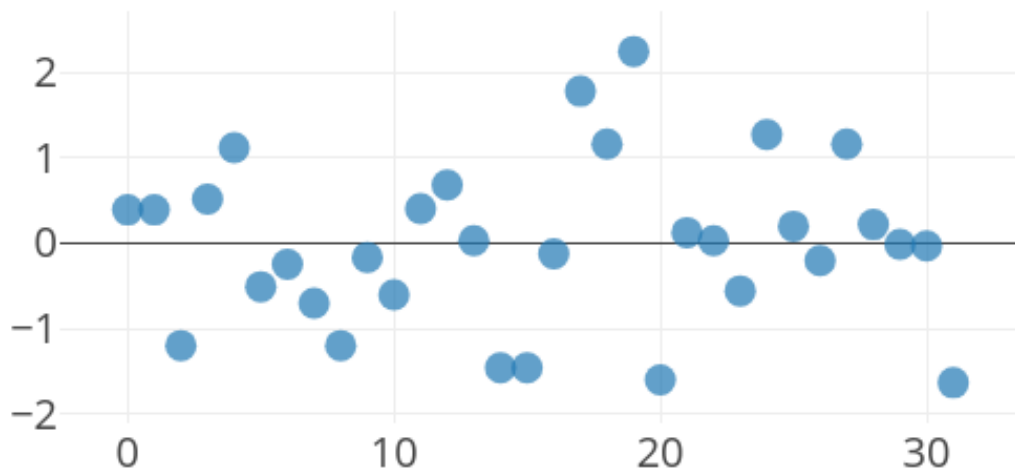
ot 006 - Scatterplot - Residuals vs Fitted.val



file:///tmp/RtmpooteVR/file58cd68ec03fc/widget58cd187e7ee7.html screenshot completed



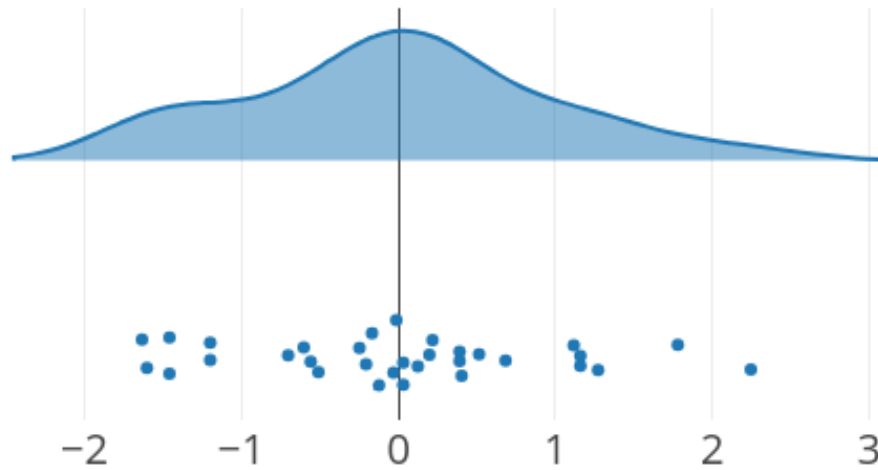
ot 007 - Scatterplot - Studentized Residuals



file:///tmp/RtmpooteVR/file58cd4af92cf1/widget58cd5a2f6fdb.html screenshot completed



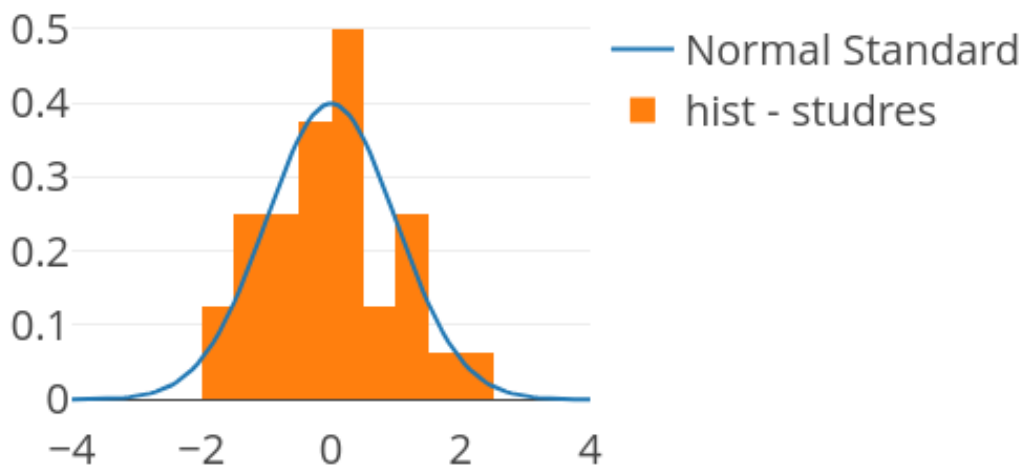
Plot 008 - Studentized Residuals



file:///tmp/RtmpooteVR/file58cd367a0f8f/widget58cd52fee9b9.html screenshot completed



Plot 009 - Studres Distribution



file:///tmp/RtmpooteVR/file58cd4ece2d49/widget58cd1d00ec9d.html screenshot completed

